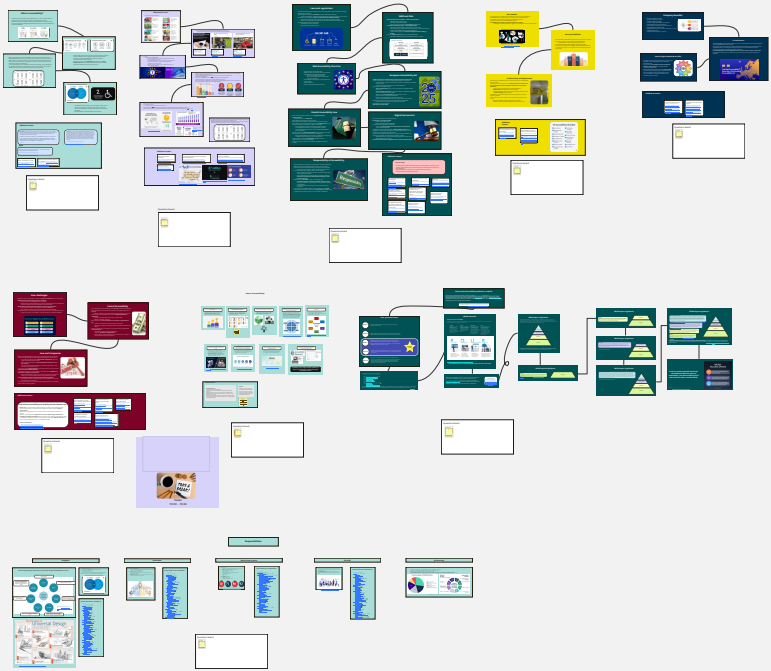
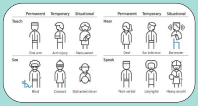

Universal Design
Universal Design
Universal Design

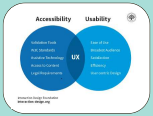
What is accessibility?

- Accessibility, often referred to by the abbreviation A11Y, is the concept of whether a product or service can be used by everyone—however they encounter it.
- Accessibility ensures that digital products, websites, applications, services and other interactive interfaces are designed and developed to be easy to use and understand by people with disabilities.

- 1.3 billion folks around the world, meaning 16% of the entire population, live with a disability and are navigating the world through assistive technology or other augmentations to enable them with their interactions in the world around them.
- Accessibility is not only the **right thing to do**, but often also **brings benefits to all users**. That's because accessibility features that help people with disabilities often help other people, too.

- While many people have permanent disabilities, every single one of us will encounter TEMPORARY disabilities as well. The sun might shine on your phone's screen so you need better contrast, you might have broken your preferred arm and need to operate things with you other hand, you might be working from a noisy location where you can't hear something so captions would help you...
- For some, handling your day-to-day tasks and issues like banking is simply impossible by doing traditional paperwork. Accessible digital services enable everyone to accomplish that and much more and empowers them in ways that nothing else can.





- The difference between usability and accessibility is that the former aims to build a great user experience while the latter specifically focuses on including the non-standard use cases and users into the mix as well.
- 80% of disabilities are invisible (ADHD, color blindness...)

Additional content

"***Accessibility*** *is the design of products, devices, services, vehicles, or environments so as to be usable by people with disabilities. The concept of accessible design and practice of accessible development ensures both "direct access" (i.e. unassisted) and "indirect access" meaning compatibility with a person's assistive technology (for example, computer screen reader).*

Accessibility can be viewed as the "ability to access" and benefit from some system or entity. The concept focuses on enabling access for people with disabilities, or enabling access through the use of assistive technology; however, research and development in accessibility brings benefits to everyone. Therefore, an accessible society should eliminate digital divide or knowledge divide."

Wikipedia

"***Accessible*** *means a person with a disability is afforded the opportunity to acquire the same information, engage in the same interactions, and enjoy the same services as a person without a disability in an equally effective and equally integrated manner, with substantially equivalent ease of use. The person with a disability must be able to obtain the information as fully, equally and independently as a person without a disability. Although this might not result in identical ease of use compared to that of persons without disabilities, it still must ensure equal opportunity to the educational benefits and opportunities afforded by the technology and equal treatment in the use of such technology."*

U.S. Department of Education's Office for Civil Rights

"*Accessibility is the practice of making information, activities, and/or environments sensible, meaningful, and usable for as many people as possible.*"

https://www.interactiveteur.com/docs/what-is-accessibility/

What is accessibility? (In Finnish): https://youtu.be/pF0Xy5DOvVE?si=xCIEDIE441Q8kz3n

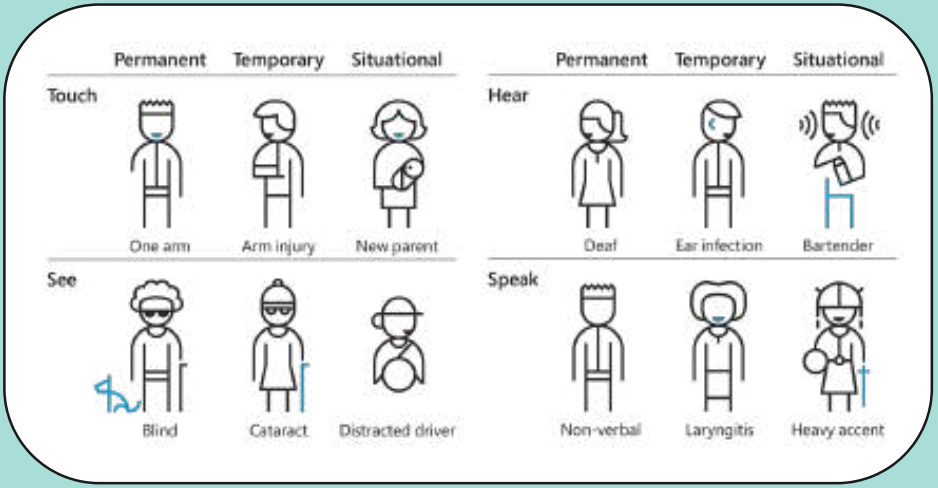
Interaction design foundation definition of accessibility: What is accessibility? (In Finnish): https://youtu.be/pF0Xy5DOvVE?si=xCIEDIE441Q8kz3n

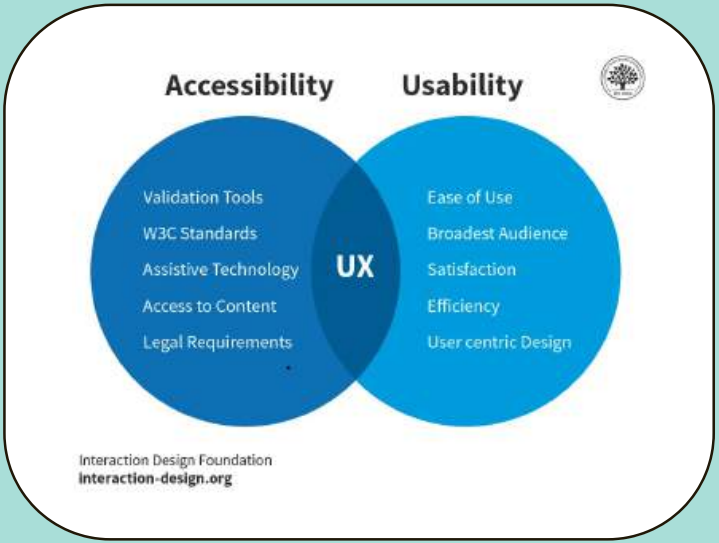
What is accessibility?

- Accessibility, often referred to by the abbreviation A11Y, is the concept of whether a product or service can be used by everyone—however they encounter it.
- Accessibility ensures that digital products, websites, applications, services and other interactive interfaces are designed and developed to be easy to use and understand by people with disabilities.

- 1.3 billion folks around the world, meaning 16% of the entire population, live with a disability and are navigating the world through assistive technology or other augmentations to enable them with their interactions in the world around them.
- Accessibility is not only the **right thing to do**, but often also **brings benefits to *all* users**. That's because accessibility features that help people with disabilities often help other people, too.

- While many people have permanent disabilities, every single one of us will encounter TEMPORARY disabilities as well. The sun might shine on your phone's screen so you need better contrast, you might have broken your preferred arm and need to operate things with you other hand, you might be working from a noisy location where you can't hear something so captions would help you...
- For some, handling your day-to-day tasks and issues like banking is simply impossible by doing traditional paperwork. Accessible digital services enable everyone to accomplish that and much more and empowers them in ways that nothing else can.





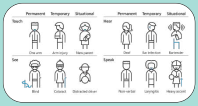
- The difference between usability and accessibility is that the former aims to build a great user experience while the latter specifically focuses on including the non-standard use cases and users into the mix as well.
- 80% of disabilities are invisible (ADHD, color blindness...)

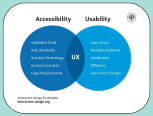
What is accessibility?

- Accessibility, often referred to by the abbreviation A11Y, is the concept of whether a product or service can be used by everyone—however they encounter it.
- Accessibility ensures that digital products, websites, applications, services and other interactive interfaces are designed and developed to be easy to use and understand by people with disabilities.

- 1.3 billion folks around the world, meaning 16% of the entire population, live with a disability and are navigating the world through assistive technology or other augmentations to enable them with their interactions in the world around them.
- Accessibility is not only the **right thing to do**, but often also **brings benefits to all users**. That's because accessibility features that help people with disabilities often help other people, too.

- While many people have permanent disabilities, every single one of us will encounter TEMPORARY disabilities as well. The sun might shine on your phone's screen so you need better contrast, you might have broken your preferred arm and need to operate things with you other hand, you might be working from a noisy location where you can't hear something so captions would help you...
- For some, handling your day-to-day tasks and issues like banking is simply impossible by doing traditional paperwork. Accessible digital services enable everyone to accomplish that and much more and empowers them in ways that nothing else can.





- The difference between usability and accessibility is that the former aims to build a great user experience while the latter specifically focuses on including the non-standard use cases and users into the mix as well.
- 80% of disabilities are invisible (ADHD, color blindness...)

Additional content

"***Accessibility*** *is the design of products, devices, services, vehicles, or environments so as to be usable by people with disabilities. The concept of accessible design and practice of accessible development ensures both "direct access" (i.e. unassisted) and "indirect access" meaning compatibility with a person's assistive technology (for example, computer screen reader).*

Accessibility can be viewed as the "ability to access" and benefit from some system or entity. The concept focuses on enabling access for people with disabilities, or enabling access through the use of assistive technology; however, research and development in accessibility brings benefits to everyone. Therefore, an accessible society should eliminate digital divide or knowledge divide."

Wikipedia

"***Accessible*** *means a person with a disability is afforded the opportunity to acquire the same information, engage in the same interactions, and enjoy the same services as a person without a disability in an equally effective and equally integrated manner, with substantially equivalent ease of use. The person with a disability must be able to obtain the information as fully, equally and independently as a person without a disability. Although this might not result in identical ease of use compared to that of persons without disabilities, it still must ensure equal opportunity to the educational benefits and opportunities afforded by the technology and equal treatment in the use of such technology."*

U.S. Department of Education's Office for Civil Rights

"*Accessibility is the practice of making information, activities, and/or environments sensible, meaningful, and usable for as many people as possible.*"

https://www.interactiveteur.com/docs/what-is-accessibility/

What is accessibility? (In Finnish): https://youtu.be/pF0Xy5DOvVE?si=xCIEDIE441Q8kz3n

Interaction design foundation definition of accessibility: What is accessibility? (In Finnish): https://youtu.be/pF0Xy5DOvVE?si=xCIEDIE441Q8kz3n

Why should I care?

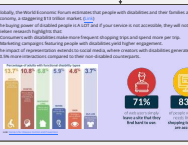
- It is humane: inclusion through design

- It is requested. Some customers demand it. This will increasingly be the case in the future.
  - NO already had a situation like this and we soon it by committing to improve accessibility

- It is the law
  - Digital services act (digipalvelulaki)
    - The service and the content must fulfill the legal minimum, if the service falls under the Digital Services Act
    - The accessibility of the service and its contents must be assessed and the status of the accessibility and its possible shortcomings must be presented in the accessibility report
    - The service should include an electronic feedback channel for users to submit accessibility feedback
    - Feedback should be answered within 14 days

- Globally, the World Economic Forum estimates that people with disabilities and their families account for 5.4% of the global economy, a staggering $13 trillion market. [Source]
- The buying power of disabled people is a LOT and if your service is not accessible, they will not use it
- Nielsen research highlights that:
  - Accessibility improvements make more frequent shopping trips and spend more per trip.
  - Marketing campaigns featuring people with disabilities yield higher engagement.

The impact of representation extends to social media, where creators with disabilities generate 21.4% more media value and 25.5% more interactions compared to their non-disabled counterparts.



- Globally, at least 2.2 billion people have a near or distance vision impairment. In at least 1 billion of these, vision impairment could have been prevented or is yet to be addressed [Source]
  - About 1 in 12 men are color blind (1 in 200 for women)
- 1.5 billion people worldwide are affected by hearing loss with 430 million of them suffering from disabling hearing loss [Source]
- More than 1 billion people are living with mental health disorders with anxiety and depressive disorders being the most common [Source]

- YOU may be affected, either now or in the future. A family member or a friend might be suffering from a disability and you need to be able to understand and help if necessary.
- Temporary disabilities happen to everyone. Broken leg doesn't care for your schedule or needs and suddenly wheelchair ramps and elevators are of the utmost importance to you.

Additional content

WCAG's introduction to accessibility:
https://www.w3.org/WAI/fundamentals/accessibility-intro/

Finland's Regional State Administrative Agency's accessibility overview:
https://www.saavutettavuus.fi/accessibility-overview/for-whom-is-accessibility-important/

Forbes on accessibility:
https://www.forbes.com/sites/forbesbusinesscouncil/2023/03/20/understanding-the-importance-of-web-accessibility/

Business side of accessibility:
https://hbr.org/2021/09/accessibility-is-a-strategy-to-consider-in-your-product-plan

Why should I care?

- It is humane: Inclusion through design

- It is requested: Some customers demand it. This will increasingly be the case in the future.
  - M2 already had a situation like this and we won it by committing to improve accessibility

Yle article about accessible controllers (in Finnish): https://yle.fi/a/74-20065107

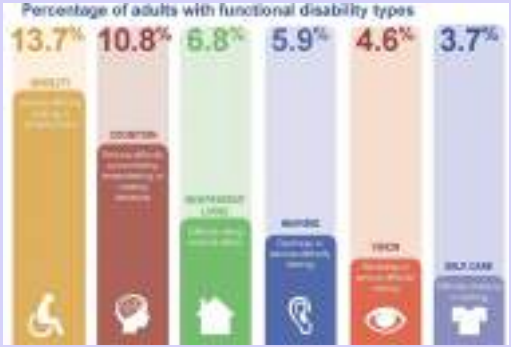
Airline passengers with disabilities speak out: https://www.bbc.com/news/articles/cjw3131x328o

Turning lives around with technology: https://www.bbc.com/news/uk-england-norfolk-68210224

- It is the law
  - Digital services act (digipalvelulaki)
    - The service and the content must fulfill the legal minimum, if the service falls under the Digital Services Act
    - The accessibility of the service and its contents must be assessed and the status of the accessibility and its possible shortcomings must be presented in the accessibility report
    - The service should include an electronic feedback channel for users to submit accessibility feedback
      - Feedback must be answered within 14 days

- Globally, the World Economic Forum estimates that people with disabilities and their families account for 54% of the global economy, a staggering $13 trillion market. (Link)
- The buying power of disabled people is A LOT and if your service is not accessible, they will not use it
- Nielsen research highlights that:
  - Consumers with disabilities make more frequent shopping trips and spend more per trip.
  - Marketing campaigns featuring people with disabilities yield higher engagement.

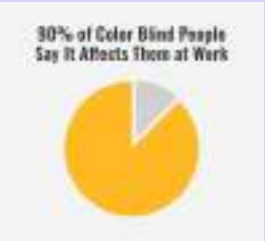
  The impact of representation extends to social media, where creators with disabilities generate 21.4% more media value and 20.5% more interactions compared to their non-disabled counterparts.

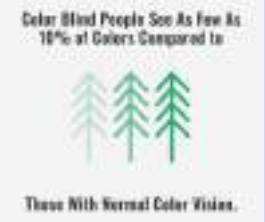


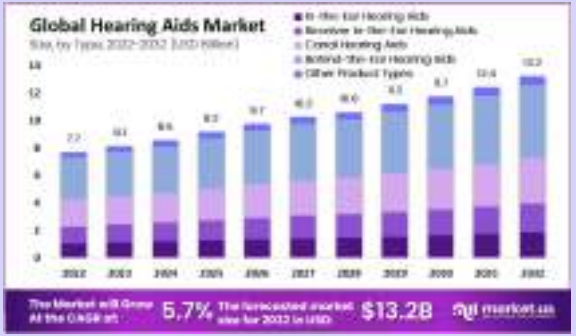
Link: Centers for Disease Control and Prevention

- Globally, at least 2.2 billion people have a near or distance vision impairment. In at least 1 billion of these, vision impairment could have been prevented or is yet to be addressed (Link)
    - About 1 in 12 men are color blind (1 in 200 for women)
- 1.5 billion people worldwide are affected by hearing loss with 430 million of them suffering from disabling hearing loss (Link)
- More than 1 billion people are living with mental health disorders with anxiety and depressive disorders being the most common (Link)

Link to color blindness images: https://enchroma.com/pages/facts-about-color-blindness



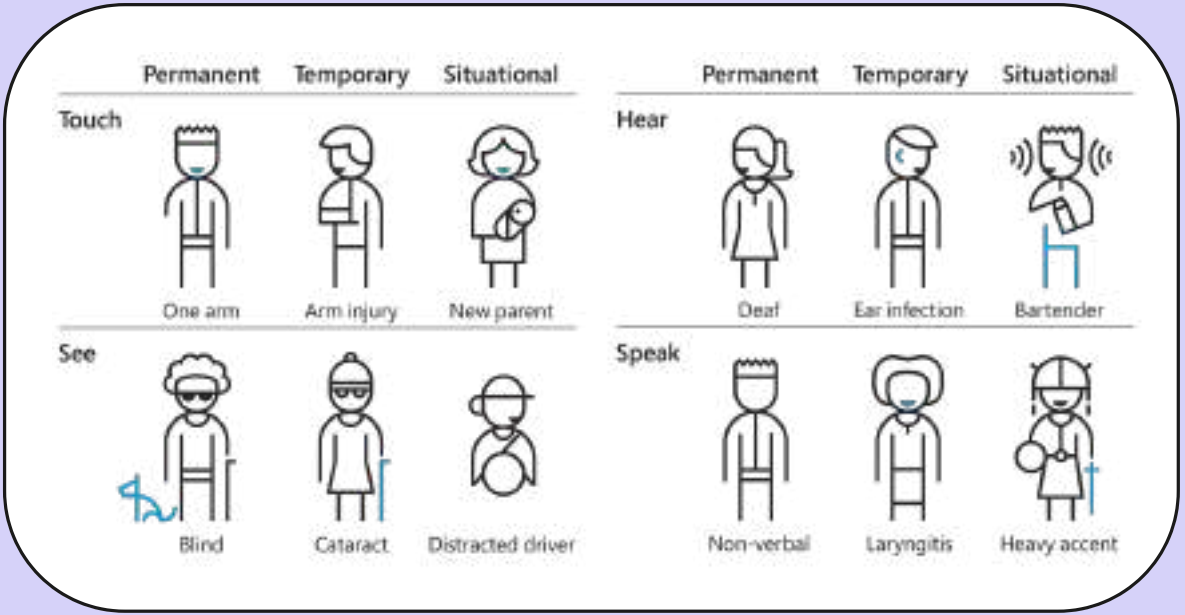




Link: https://media.market.us/hearing-loss-statistics/

Link: https://iris.who.int/server/api/core/bitstreams/31714489-1345-4439-8b37-6cbdc52e15ca/content

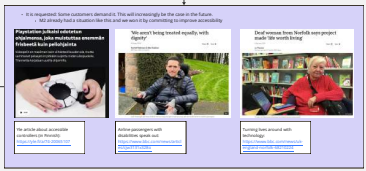
- YOU may be affected, either now or in the future. A family member or a friend might be suffering from a disability and you need to be able to understand and help if necessary
- Temporary disabilities happen to EVERYONE. A broken leg doesn't care for your schedule or needs and suddenly wheelchair ramps and elevators are of the utmost importance to you



Why should I care?

- It is humane. Inclusion through design

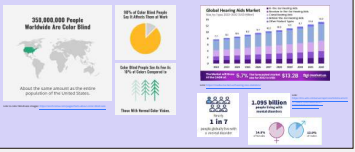
- It is required. Some customers demand it. This will increasingly be the case in the future.
  - MZ already had a situation like this and we won it by committing to improve accessibility



- It is the law
  - Digital services act (gigaparlakuki)
    - The service and the content must fulfill the legal minimum, if the service falls under the Digital Services Act
    - The accessibility of the service and its contents must be assessed and the status of the accessibility and its possible shortcomings must be presented in the accessibility report
    - The service should include an electronic feedback channel for users to submit accessibility feedback
      - Feedback must be answered within 14 days

- Globally, the World Economic Forum estimates that people with disabilities and their families account for 54% of the global economy, a staggering $13 trillion market. (Link)
- The buying power of disabled people is A LOT bad if your service is not accessible, they will not use it
- Nielsen research highlights that:
  - Consumers with disabilities make more frequent shopping trips and spend more per trip.
  - Marketing campaigns featuring people with disabilities yield higher engagement.
  - The impact of representation extends to social media, where creators with disabilities generate 21.6% more media value and 26.3% more interactions compared to their non-disabled counterparts.

- Globally, at least 2.2 billion people have a near or distance vision impairment. In at least 1 billion of these, vision impairment could have been prevented or is yet to be addressed (Link)
  - About 1 in 12 men (8% are color blind) in 200 for women
- 1.5 billion people worldwide are affected by hearing loss with 430 million of them suffering from disabling hearing loss (Link)
- More than 1 billion people are living with mental health disorders with anxiety and depressive disorders being the most common (Link)



- YOU may be affected, either now or in the future. A family member or a friend might be suffering from a disability and you need to be able to understand and help if necessary.
- Temporary disabilities happen to EVERYONE. A broken leg doesn't care for your schedule or needs and suddenly wheelchair ramps and elevators are of the utmost importance to you

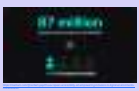
Additional content

WCAG's introduction to accessibility:
https://www.w3.org/WAI/fundamentals/accessibility-intro/

Finland's Regional State Administrative Agency's *accessibility overview*:
https://www.webaccessibility.fi/accessibility-overview/for-whom-is-accessibility-important/

Forbes on accessibility:
https://www.forbes.com/sites/forbesbusinesscouncil/2023/03/20/understanding-the-importance-of-web-accessibility/

Business side of accessibility:
https://techjunkie.com/accessibility-web-crunching-numbers-accessibility/



Laws and regulations

- The basis for the demands on Accessibility lies in laws and regulations which have been increasingly demanding in recent years.
- In EU the regulation is based on **EN 301 549**, a Harmonized European Standard
  - Harmonized Standard = Expected to be implemented as accurately as possible, as national law
- EN 301 549 is currently based on **WCAG 2.1 (Web Content Accessibility Guidelines)**
  - Also has parts referring to other areas such as documentation and user settings
  - WCAG 2.2 was published in 2020 however, EN 301 549 is expected to amend it first 2024.
- Two general obligations for member states:
  - Possibility for consumer complaints
  - Enforcement by national supervisors

WAD and EAA

- **Web Accessibility Directive (WAD) - "Saavutettavuusdirektiivi"**
  - Derived directly from EN 301 549
  - Specifically used for public entities
  - Enforced in most member states since **2020** including Finland
- **European Accessibility Act (EAA) - "Esteettömyysdirektiivi"**
  - Specifically for private entities in Business to Consumer sector
  - EAA is originally not created from EN 301 549, however is now been amended to conform towards WAD and EN 301 549 (and WCAG 2.1)
  - Implementation began 2022, **enforcement Begun 28.6.2025, including in Finland**
- In Finland, **Digital Services Act** covers now both WAD and EAA
  - Originally only WAD, however it was amended in 2022 to cover both

Web Accessibility Directive

- Derived directly from the EN 301 549
- Obligee all public sector bodies in the EU to make their online websites and mobile apps accessible
  - Governments, public companies etc.
  - Also, critical private entities, such banks and insurance companies
- Online websites and applications, mobile applications, documentation
- Implemented 2019
- Enforcement began 2020 (2021 for mobile)

European Accessibility Act

- **Applies to Business to Consumer (B2C) products and services sold in the EU**
- Covers a large selection of consumer targeted digital appliances and services
  - All services where "a contract, for products or services, is created are considered to be online services"
  - Requirements apply to both online- and mobile applications
- Key exceptions on product coverage
  - **Does not include B2B** products or services marketed and targeted towards consumers
  - **Does not include End of Life** products (No updates, no maintenance, if offered to customers deficiencies must be stated)
  - Does not include **micro-enterprises** (<10 employees, <2 mil. €)
  - May not turn into a disproportionate burden
  - Does not require fundamental alterations to a product
  - Certain pre-recorded media, archives, third party content are also exempt
- Many existing products and services have a grace period until 28.6.2030
  - Physical services in particular, such as self-service terminals
- Implementation started 28.6.2022
- **Enforcement starts 28.6.2025**

Finnish Accessibility Law

- The core of Finnish legal accessibility is on the **Digital Services Act ("Digipalvelulaki")**
  - Based on EU directive **EN 301 549**
- Covers both private and public sectors in Finland
- Public entities must comply with requirements from 3 other laws as well:
- Act on Public Procurement and Concession Contracts (**Hankintalaki**) requires taking accessibility into consideration in public procurements
  - Public organizations also often determine the required level of accessibility themselves and it is not always WCAG compliance
  - This means that the requirements may vary wildly between different public organizations
- Non-discrimination Act (**Yhdenvertaisuuslaki**) obligates employers to ensure that employees can perform their duties even with disabilities
  - Does not demand features to be accessible - adaptations are enough, e.g. a disabled person's supervisor can input working hours
- Administrative Procedures Act (**Hallintolaki**) obligates the use of understandable, correct and clear language
- Traficom is the current supervisor of accessibility of digital services (since 1.1.2020)

Digital Services Act

- The primary Finnish Law concerning Accessibility
- **Passed 2019** and **included only WAD**
  - Covered only public domain
  - Enforcement began 2020 (mobile 2021)
- In 2022 the law was amended to include EAA
  - Covers now **both the public sector and contain private enterprises**
  - Implementation began 2022
- The Digital Services Act refers specifically to **WAD and EAA directives** and is designed to empower government officials, stating their legal powers, as well as requirements for companies and public entities
- Follows very closely EN 301 549 in application, having the same exceptions and requirements, however, lacks exact level of required accessibility
  - Demands ease of use, without actually specifying what this actually means
  - Shifts the task to the supervisor
- **Private sector enforcement, in applicable cases, is stated to begin 28.6.2025**

Responsibility of Accessibility

- The responsibility for accessibility in products works somewhat unintuitively
- When a product is under the Digital Services Act, the service provider is responsible for the accessibility, even when the service is from a 3rd party
  - This extends to all 3rd party modules, software, apps and so forth
- This causes both private and public spheres to act proactively
  - Private companies must invest in vendor cooperation and be careful with using 3rd party software
  - Public services demand increasingly higher accessibility in offerings
  - **Public services are usually the main method of ensuring the correct level of accessibility**
- Software vendor, **Iko Viona**, in Espoo is a strange situation
  - **Law technically does not demand accessibility directly from us**
  - However, the customer may be subject to the consequences specified in the legislation
  - Thus the customer has a pressure to remove products with poor accessibility
- As a conclusion, as a software vendor, Viona (in Finland) is rarely **legally** responsible for accessibility, however we are **contractually** obligated to fulfill accessibility which is often the basis of our accessibility requirements.

Additional content

Finnish Oversight

- Current supervising official in Finland is Traficom. Previously (before 1.1.2020) it was AVI.
- Traficom makes evaluation tours automatically as well as manual checks to desktop and mobile applications
- They also handle any complaints citizens might send about Accessibility
- So far (Q1/2023) enforcement has been light

Web Accessibility Directive (WAD - EU2016/2102) 2:
https://digital-strategy.ec.europa.eu/en/policies/web-accessibility-directive-standards-and

European Accessibility Act (EAA - EU2019/882) 3:
https://ec.europa.eu/social/main.jsp?catId=1202

EN 301 549:
https://www.etsi.org/deliver/etsi_en/301549_301549v020101p.pdf?ref=https://www.etsi.org/deliver/etsi_en/301549_301549v020101p.pdf

Laki digitaalisten palvelujen tarjoamisesta ja sen muuttamisesta (EAA amendment to Finnish law):
https://www.finlex.fi/fi/laki/alkup/2022/20220750

Laki digitaalisten palvelujen tarjoamisesta (Finnish law concerning providing digital services):
https://www.finlex.fi/fi/laki/alkup/2019/20190306

European Union 10-year Accessibility Strategy
https://ec.europa.eu/social/main.jsp?catId=1484&langId=en

Laki digitaalisten palvelujen tarjoamisesta annetun lain muuttamisesta (EAA amendment to Finnish law):
https://www.finlex.fi/fi/laki/alkup/2022/20220750

European Commission Accessibility Centre
https://ec.europa.eu/info/index_en

Laws and regulations

- The basis for the demands on Accessibility lies in laws and regulations which have been Increasingly demanding in recent years.
- In EU the legislation is based on **EN 301 549**, a Harmonized European Standard
    - Harmonized Standard = Expected to be implemented as accurately as possible, as national law
- EN 301 549 is currently based on **WCAG 2.1 (Web Content Accessibility Guidelines)**
    - Also has parts referring to other areas such as documentation and user settings
    - WCAG 2.2 was published in 2025 however, EN 301 549 is expected to amend it first 2026
- Two general obligations for member states:
    - Possibility for consumer complaints
    - Enforcement by national supervisors

WAD and EAA

- **Web Accessibility Directive (WAD) - "Saavutettavuusdirektiivi"**
  - Derived directly from EN 301 549
  - Specifically used for public entities
  - Enforced in most member states **since 2020** including Finland
- **European Accessibility Act (EAA) - "Esteettömyysdirektiivi"**
  - Specifically for private entities in Business to Consumer sector
  - EAA is originally not created from EN 301 549, however is now been amended to conform towards WAD and EN 301 549 (and WCAG 2.1)
  - Implementation began 2022, **enforcement begun 28.6.2025, including in Finland.**
- In Finland, **Digital Services Act** covers now both WAD and EAA.
  - Originally only WAD, however it was amended in 2023 to cover both

Web Accessibility Directive

- Derived directly from the EN 301 549
- Obliges all public sector bodies in the EU to make their online websites and mobile apps accessible
    - Governments, public companies etc.
    - Also, critical private entities, such banks and insurance companies
- Online websites and applications, mobile applications, documentation
- Implemented 2019
- Enforcement began 2020 (2021 for mobile)

European Accessibility Act

- **Applies to Business to Consumer (B2C) products and services sold in the EU**
- Covers a large selection of consumer targeted digital appliances and services
  - All services where "a contract, for products or services, is created are considered to be online stores"
  - Requirements apply to both online- and mobile applications
- Key exceptions on product coverage
  - **Does not include B2B** products or services marketed and targeted towards consumers
  - **Does not include End of Life** products (No updates, no maintenance, if offered to customers deficiencies must be stated)
  - Does not include **micro-enterprises** (>10 employees, >2 mil. €)
  - May not turn into a disproportionate burden
  - Does not require fundamental alterations to a product
  - Certain pre-recorded media, archives, third party content are also exempt
- Many existing products and services have a grace period until 28.6.2030
  - Physical services in particular, such as self-service terminals
- Implementation started 28.6.2022
- **Enforcement starts 28.6.2025**

Finnish Accessibility Law

- The core of Finnish legal accessibility is on the **Digital Services Act ("Digipalvelulaki")**
  - Based on EU directive **EN 301 549**
- Covers both private and public sectors in Finland
- Public entities must comply with requirements from 3 other laws as well:
- Act on Public Procurement and Concession Contracts (**Hankintalaki)** requires taking accessibility into consideration in public procurements
  - Public organizations also often determine the required level of accessibility themselves and it is not always WCAG compliance.
  - This means that the requirements may vary wildly between different public organizations
- Non-discrimination Act (**Yhdenvertaisuuslaki**) obligates employers to ensure that employees can perform their duties even with disabilities
  - Does not demand features to be accessible - adaptations are enough, e.g. a disabled person's supervisor can input working hours
- Administrative Procedures Act (**Hallintolaki**) obligates the use of understandable, correct and clear language
- Traficom is the current supervisor of accessibility of digital services (since 1.1.2025)

Digital Services Act

- The primary Finnish Law concerning Accessibility
- **Passed 2019** and **included only WAD**
  - Covered only public domain
  - Enforcement began 2020 (mobile 2021)
- In 2023 the law was amended to include EAA
  - Covers now **both the public sector and certain private enterprises**
  - Implementation began 2022
- The Digital Services Act refers specifically to **WAD and EAA directives** and is designed to empower government officials, stating their legal powers, as well as requirements for companies and public entities
- Follows very closely EN 301 549 in application, having the same exceptions and requirements, however, lacks exact level of required accessibility
  - Demands ease of use, without actually specifying what this actually means
  - Shifts the task to the supervisor
- **Private sector enforcement, in applicable cases, is slated to begin 28.6.2025**

Responsibility of Accessibility

- The responsibility for accessibility in products works somewhat unintuitively
- When a product is under the Digital Services Act, the service provider is responsible for the accessibility, even when the service is from a 3rd party
    - This extends to all 3rd party modules, software, apps and so forth
- This causes both private and public spheres to act proactively:
    - Private companies must invest in vendor cooperation and be careful with using 3rd party software
    - Public entities demand increasingly higher accessibility in offerings
    - **Contracts are usually the main method of ensuring the correct level of accessibility**
- Software vendors**, like Visma**, end up in a strange situation
    - **Law technically does not demand accessibility directly from us**
    - However, the customer may be subject to the consequences specified in the legislation
    - Thus the customer has a pressure to remove products with poor accessibility
- As a conclusion: as a software vendor, Visma (in Finland) is rarely **legally** responsible for accessibility, however we are **contractually** obligated to fulfill accessibility which is often the basis of our accessibility requirements.

Laws and regulations

- The basis for the demands on Accessibility lies in laws and regulations which have been increasingly demanding in recent years.
- In EU the legislation is based on **EN 301 549**, a Harmonized European Standard
  - Harmonized Standard = Expected to be implemented as accurately as possible, as national law
- EN 301 549 is currently based on **WCAG 2.1 (Web Content Accessibility Guidelines)**
  - Also has parts referring to other areas such as documentation and user settings
  - WCAG 2.2 was published in 2023 however, EN 301 549 is expected to amend it first 2026
- Their general obligations for member states:
  - Possibility for consumer complaints
  - Enforcement by national supervisors

WAD and EAA

- **Web Accessibility Directive (WAD) - "Saavutettavuusdirektiivi"**
  - Derived directly from EN 301 549
  - Specifically used for public entities
  - Enforced in most member states **since 2020** including Finland
- **European Accessibility Act (EAA) - "Esteettömyysdirektiivi"**
  - Specifically for private entities in Business to Consumer sector
  - EAA is originally not created from EN 301 549, however is now been amended to conform towards WAD and EN 301 549 (and WCAG 2.1)
  - Implementation began 2022, **enforcement began 28.6.2025, including in Finland.**
- In Finland, **Digital Services Act** covers now both WAD and EAA.
  - Originally only WAD, however it was amended in 2023 to cover both

Web Accessibility Directive

- Derived directly from the EN 301 549
- Obliges all public sector bodies in the EU to make their online websites and mobile apps accessible
  - Governments, public companies etc.
  - Also, critical private entities, such banks and insurance companies
- Online websites and applications, mobile applications, documentation
- Implemented 2019
- Enforcement began 2020 (2021 for mobile)

European Accessibility Act

- **Applies to Business to Consumer (B2C) products and services sold in the EU**
- Covers a large selection of consumer targeted digital appliances and services
  - All services where "a contract, for products or services, is created are considered to be online stores"
  - Requirements apply to both online- and mobile applications
- Key exceptions on product coverage
  - **Does not include B2B** products or services marketed and targeted towards consumers
  - **Does not include End of Life** products (No updates, no maintenance, if offered to customers deficiencies must be stated)
  - Does not include **micro-enterprises** (< 10 employees, <2 mil. €)
    - May not turn into a disproportionate burden
  - Does not require fundamental alterations to a product
  - Certain pre-recorded media, archives, third party content are also exempt
- Many existing products and services have a grace period until 28.6.2030
  - Physical services in particular, such as self-service terminals
- Implementation started 28.6.2022
- **Enforcement starts 28.6.2025**

Finnish Accessibility Law

- The core of Finnish legal accessibility is on the **Digital Services Act ("Digipalvelulaki")**
  - Based on EU directive **EN 301 549**
- Covers both private and public sectors in Finland
- Public entities must comply with requirements from 3 other laws as well:
- Act on Public Procurement and Concession Contracts (**Hankintalaki**) requires taking accessibility into consideration in public procurements
  - Public organizations also often determine the required level of accessibility themselves and it is not always WCAG compliance.
  - This means that the requirements may vary widely between different public organizations
- Non-discrimination Act (**Yhdenvertaisuuslaki**) obligates employers to ensure that employees can perform their duties even with disabilities
  - Does not demand features to be accessible - adaptations are enough, e.g. a disabled person's supervisor can input working hours
- Administrative Procedures Act (**Hallintolaki**) obligates the use of understandable, correct and clear language
- Traficom is the current supervisor of accessibility of digital services (since 1.1.2025)

Digital Services Act

- The primary Finnish Law concerning Accessibility
- **Passed 2019** and **included only WAD**
  - Covered only public domain
  - Enforcement began 2020 (mobile 2021)
- In 2023 the law was amended to include EAA
  - Covers now **both the public sector and certain private enterprises**
  - Implementation began 2022
- The Digital Services Act refers specifically to **WAD and EAA directives** and is designed to empower government officials, stating their legal powers, as well as requirements for companies and public entities
- Follows very closely EN 301 549 in application, having the same exceptions and requirements, however, lacks exact level of required accessibility
  - Demands ease of use, without actually specifying what this actually means
  - Shifts the task to the supervisor
- **Private sector enforcement, in applicable cases, is slated to begin 28.6.2025**

Responsibility of Accessibility

- The responsibility for accessibility in products works somewhat unintuitively
- When a product is under the Digital Services Act, the service provider is responsible for the accessibility, even when the service is from a 3rd party
  - This extends to all 3rd party modules, software, apps and so forth
- This causes both private and public sphere to act proactively:
  - Private companies must invest in vendor cooperation and be careful with using 3rd party software
  - Public entities demand increasingly higher accessibility in offerings
  - **Contracts are usually the main method of ensuring the correct level of accessibility**
- Software vendors, **like Visma**, end up in a strange situation
  - **Law technically does not demand accessibility directly from us**
  - However, the customer may be subject to the consequences specified in the legislation
  - Thus the customer has a pressure to remove products with poor accessibility
- As a producer/seller of software vendor, Visma (in Finland) is rarely **legally** responsible for accessibility, however we are **contractually** obligated to fulfill accessibility which is often the basis of our accessibility requirements.

Additional content

Finnish Oversight

- Current supervising official in Finland is Traficom. Previously (before 1.1.2025) it was AVI.
- Traficom makes evaluation tours automatically as well as manual checks to desktop and mobile applications
- They also handle any complaints citizens might send about Accessibility
- So far (Q1/2025) enforcement has been light

Web Accessibility Directive (WAD - EU2016/2102):
https://digital-strategy.ec.europa.eu/en/policies/web-accessibility-directive-implementation-and-monitoring

European Accessibility Act (EAA - EU2019/882):
https://hse.europa.eu/social/main.jsp?catId=1202

EN 301 549:
https://www.etsi.org/deliver/etsi_en/301500_301599/301549/03.02.01_60/en_301549v030201p.pdf

Laki eräiden tuotteiden esteettömyysvaatimuksista:
https://www.finlex.fi/fi/laki/alkup/2022/20220102

Laki digipalvelun palvelujen (esteettömyysvaatimuksista a.k.a "Digipalvelulaki" (Finnish law concerning providing digital services):
https://www.finlex.fi/fi/laki/alkup/2019/20190306

European Union 10-year Accessibility Strategy:
https://ec.europa.eu/social/main.jsp?catId=1484&langId=en

Laki digipalvelun palvelujen (esteettömyysvaatimuksista annetun lain muutamisesta EAA amendment to Finnish law):
https://www.finlex.fi/fi/laki/alkup/2022/20220102

European Commission Accessibility Centre
https://accessibility-eu-centre.ec.europa.eu/index_en

User benefits

- As a general rule, accessibility improves usability for everyone, not just "disabled"
- Accessibility enables a larger user pool, as it simply aims to include everyone
- Independence for every user and group regardless of the situation
- Growing commitment and trust towards the company
- **Accessibility does not bring any negative impacts to a product**
  - All features Accessibility are also for regular users and makes life easier for them as well

Image source: https://www.freepik.com/search?dl=1&format=png&size=medium&illustrations&kw=accessibility&page=1&sortby=relevance&type=illustrations

Not only disabilities

- Most accessibility related features are general improvements on usability
- Recognition, instead of memorization, reduce cognitive load
  - Standardized icons and standardized layout practices
- Correct use of contrast make everything easier to read and use
- Accessibility improvements in navigation, readability and well organized content improve retention rates, reduce bounce rates and improves the quality of life

Preferred way of doing business

- A wide selection of disabilities can turn common services into complicated and difficult ventures
- Accessible online services and products **enables customers to interact with services completely without any physical interaction or direct human interaction**
  - Hearing, speech and sight are all complicating disabilities
  - **Social anxiety,** several different mental conditions
  - Cognitive difficulties cause the user to prefer taking time with decisions and **prefer reading** about the subject
- Accessible sites and services are faster, but also cheaper as less customer support is required
- Turning services into online services remove requirements physically visit a venue
  - Difficult commutes or restricting open hours complicate
  - Movement related disabilities can significantly complicate visits

Additional content

6 Unexpected Benefits of Web Accessibility:
https://www.boia.org/blog/6-unexpected-benefits-of-web-accessibility

Why Accessibility Is Critical in Custom Software Development
https://www.taazaa.com/software-accessibility/#:~:text=Accessibility%20ensures%20that%20individuals%20with,users%2C%20regardless%20of%20their%20abilities.

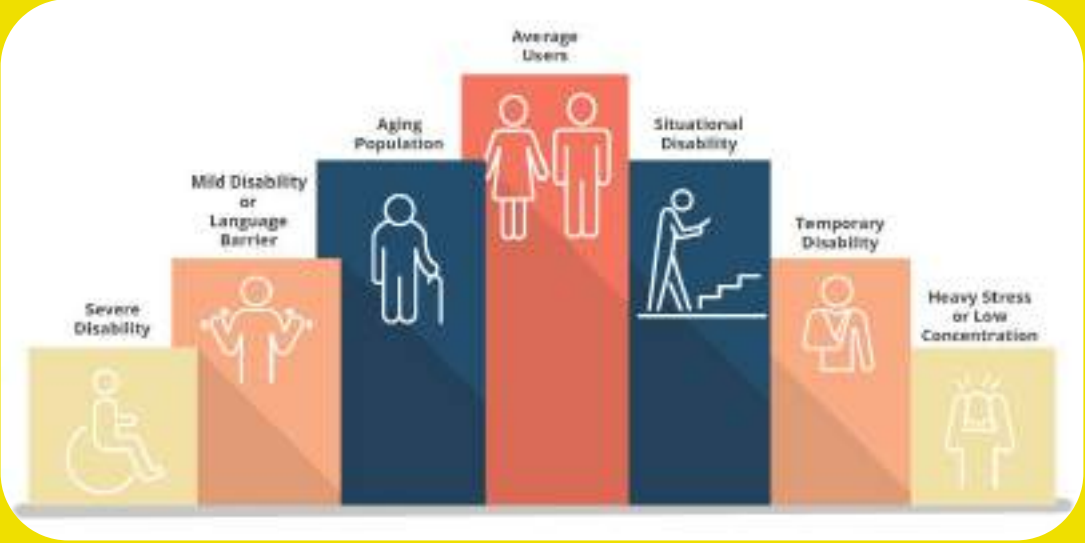
User benefits

- As a general rule, accessibility improves usability for everyone, not just "disabled"
- Accessibility enables a larger user pool, as it simply aims to include everyone
- Independence for every user and group regardless of the situation
- Growing commitment and trust towards the company
- **Accessibility does not bring any negative impacts to a product**
    - All features Accessibility are also for regular users and makes life easier for them as well

Image source: https://medium.com/@sonakshi.arora02/empowering-inclusion-leveraging-artificial-intelligence-for-improving-accessibility-27cf17b28c54

Not only disabilities

- Most accessibility related features are general improvements on usability
- Recognition, instead of memorization, reduce cognitive load
    - Standardized icons and standardized layout practices
- Correct use of contrast make everything easier to read and use
- Accessibility improvements in navigation, readability and well organized content improve retention rates, reduce bounce rates and improves the quality of life



Preferred way of doing business

- A wide selection of disabilities can turn common services into complicated and difficult ventures
- Accessible online services and products **enables customers to interact with services completely without any physical interaction or direct human interaction**
    - Hearing, speech and sight are all complicating disabilities
    - **Social anxiety,** several different mental conditions
    - Cognitive difficulties cause the user to prefer taking time with decisions and **prefer reading** about the subject
- Accessible sites and services are faster, but also cheaper as less customer support is required
- Turning services into online services remove requirements physically visit a venue
    - Difficult commutes or restricting open hours complicate
    - Movement related disabilities can significantly complicate visits

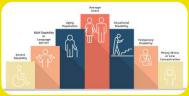
User benefits

- As a general rule, accessibility improves usability for everyone, not just "disabled"
- Accessibility enables a larger user pool, as it simply aims to include everyone
- Independence for every user and group regardless of the situation
- Growing commitment and trust towards the company
- **Accessibility does not bring any negative impacts to a product**
  - All features Accessibility are also for regular users and makes life easier for them as well

Image source: https://www.boia.org/blog/6-unexpected-benefits-of-web-accessibility
https://www.taazaa.com/software-accessibility/#:~:text=Accessibility%20ensures%20that%20individuals%20with,users%2C%20regardless%20of%20their%20abilities.

Not only disabilities

- Most accessibility related features are general improvements on usability
- Recognition, instead of memorization, reduce cognitive load
  - Standardized icons and standardized layout practices
- Correct use of contrast make everything easier to read and use
- Accessibility improvements in navigation, readability and well organized content improve retention rates, reduce bounce rates and improves the quality of life



Preferred way of doing business

- A wide selection of disabilities can turn common services into complicated and difficult ventures
- Accessible online services and products **enables customers to interact with services completely without any physical interaction or direct human interaction**
  - Hearing, speech and sight are all complicating disabilities
  - **Social anxiety,** several different mental conditions
  - Cognitive difficulties cause the user to prefer taking time with decisions and **prefer reading** about the subject
- Accessible sites and services are faster, but also cheaper as less customer support is required
- Turning services into online services remove requirements physically visit a venue
  - Difficult commutes or restricting open hours complicate
  - Movement related disabilities can significantly complicate visits

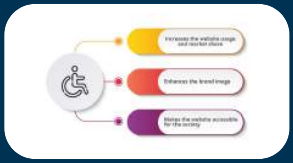
Additional content

6 Unexpected Benefits of Web Accessibility:
https://www.boia.org/blog/6-unexpected-benefits-of-web-accessibility

Why Accessibility Is Critical in Custom Software Development
https://www.taazaa.com/software-accessibility/#:~:text=Accessibility%20ensures%20that%20individuals%20with,users%2C%20regardless%20of%20their%20abilities.

Company benefits

- No loss of potential customers
- Improves usability for everyone
- Notable marketing advantage
- Humane image is projected for the customers
- Saves development time and resources
- Simple solutions, easy to maintain, less effort to fix
- Improved SEO from semantic HTML
- Opportunities to reach more users on more devices, in more settings/environments
- Enhanced public image for your brand



It is the future

- EU has been **very** proactive with accessibility for well over a decade and has consistently brought the rights of disabled people to the forefront
- EU legislation has continuously been evolved towards greater accessibility and currently the EU has dedicated itself for a second 10-year-strategy to bring online services accessible for everyone
- Thus the requirements are expected to tighten in the future
- **As of 28.6.2025 private services targeted towards consumers need to conform to EN 301 549**
- Outside EU the development is similarly evolving, however, they rarely are as demanding as EU legislation

Search Engine Optimization (SEO)

- An added bonus for having clear structures, semantics, tags and tittles is Search Engine Optimization, which gives benefits two ways:
- First benefit is that **finding desired content inside the website is simply easier,** as everything is labeled consistently and effectively
- Second is that search engines, like Google, **find your site** more efficiently by ranking it higher in algorithms, mainly due to the algorithm paying special attention to tags, structure, headers etc: as in, the same details that constitutes good accessibility

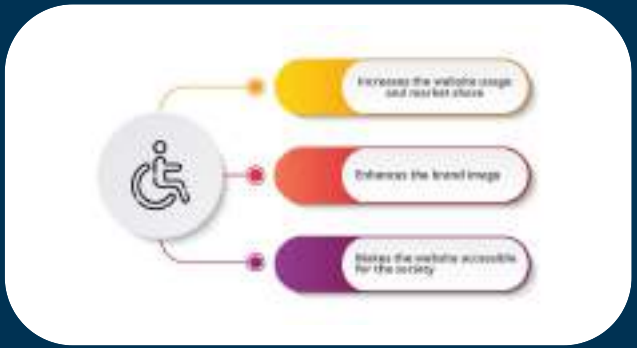
Additional content

Coding accessibility principles
What is accessible code? 6 Unexpected Benefits of Web Accessibility: https://www.boia.org/blog/6-unexpected-benefits-of-web-accessibility

How SEO and Accessibility Work Together
Coding accessibility principles
What is accessible code? 6 Unexpected Benefits of Web Accessibility: https://www.boia.org/blog/6-unexpected-benefits-of-web-accessibility

Company benefits

- No loss of potential customers
- Improves usability for everyone
- Notable marketing advantage
- Humane image is projected for the customers
- Saves development time and resources
- Simple solutions, easy to maintain, less effort to fix
- Improved SEO from semantic HTML
- Opportunities to reach more users on more devices, in more settings/environments
- Enhanced public image for your brand



It is the future

- EU has been **very** proactive with accessibility for well over a decade and has consistently brought the rights of disabled people to the forefront
- EU legislation has continuously been evolved towards greater accessibility and currently the EU has dedicated itself for a second 10-year-strategy to bring online services accessible for everyone
- Thus the requirements are expected to tighten in the future
- **As of 28.6.2025 private services targeted towards consumers need to conform to EN 301 549**
- Outside EU the development is similarly evolving, however, they rarely are as demanding as EU legislation

Search Engine Optimization (SEO)

- An added bonus for having clear structures, semantics, tags and tittles is Search Engine Optimization, which gives benefits two ways:
- First benefit is that **finding desired content inside the website is simply easier,** as everything is labeled consistently and effectively
- Second is that search engines, like Google, **find your site** more efficiently by ranking it higher in algorithms, mainly due to the algorithm paying special attention to tags, structure, headers etc: as in, the same details that constitutes good accessibility

Company benefits

- No loss of potential customers
- Improves usability for everyone
- Notable marketing advantage
- Humane image is projected for the customers
- Saves development time and resources
- Simple solutions, easy to maintain, less effort to fix
- Improved SEO from semantic HTML
- Opportunities to reach more users on more devices, in more settings/environments
- Enhanced public image for your brand

It is the future

- EU has been **very** proactive with accessibility for well over a decade and has consistently brought the rights of disabled people to the forefront
- EU legislation has continuously been evolved towards greater accessibility and currently the EU has dedicated itself for a second 10-year-strategy to bring online services accessible for everyone
- Thus the requirements are expected to tighten in the future
- **As of 28.6.2025 private services targeted towards consumers need to conform to EN 301 549**
- Outside EU the development is similarly evolving, however, they rarely are as demanding as EU legislation

Search Engine Optimization (SEO)

- An added bonus for having clear structures, semantics, tags and tittles is Search Engine Optimization, which gives benefits two ways:
- First benefit is that **finding desired content inside the website is simply easier,** as everything is labeled consistently and effectively
- Second is that search engines, like Google, **find your site** more efficiently by ranking it higher in algorithms, mainly due to the algorithm paying special attention to tags, structure, headers etc: as in, the same details that constitutes good accessibility

Additional content

Coding accessibility principles
What is accessible code? 6 Unexpected Benefits of Web Accessibility: https://www.boia.org/blog/6-unexpected-benefits-of-web-accessibility

How SEO and Accessibility Work Together
Coding accessibility principles
What is accessible code? 6 Unexpected Benefits of Web Accessibility: https://www.boia.org/blog/6-unexpected-benefits-of-web-accessibility

User challenges

- Deficiencies in basic Accessibility link directly to **loss of user retention** and more importantly **disables certain users from using a product completely**
  - Screen reader users and keyboard users in particular
  - Most features that enable these users on a site are relatively simple and cost effective, as long as the necessary steps are embedded into the product early
- **Complicated user interfaces disincentivizes users from using your website**,
  - Users simply do not want to use a difficult or complicated websites
  - Due to internet, there is likely to be a more pleasant experience available
- The high speed nature of modern way of life drives people to use fast, uncomplicated services
  - The decision to buy or use a website is less likely to happen on complicated websites

Cost of Accessibility

- The central question with accessibility is often **the cost** involved with accessibility
- When accessibility is implemented into products from the beginning the cost is negligible
  - Mainly a question of good work methodology and good design principles
- Implementing accessibility afterwards is however **always expensive and complicated**
  - Legacy code, unexpected interconnections, product bloat
  - Rewrites, redesigns, refactoring
- Bad accessibility cause **loss of revenue**, **as the customers simply refuse to buy**
  - Customer responsible for the accessibility in their product
  - Public sector companies and entities require accessibility standards to be fulfilled
- Sanctions, while unlikely to be placed on our products directly, will affect **our customers instead**
  - The effect on us would be equally devastating
  - Customer **very likely to change product**
  - If our product is mentioned to be the cause of the sanctions, the public relations fallout is likely to be significant

Law and Companies

- Most accessibility legislation is based on the harmonized EU directive, EN 301 549, therefore the national legislation might differ between different member states
  - For instance, the basis and size of **enforcement and penalties** might vary significantly between member nations
  - Thus a single accessibility issue **can be penalized several times, once in every country**
  - This also means that the responsibility of accessibility might differ in other countries from the Finnish version
- In Finland, the law is **Digital Services Act**, which states that the sanction falls under **Penalty Payment Act (Uhkasakkolaki)**
- If a sanction is placed on a product, the fine is calculated with the **capabilities of the product's company, not the product.**
  - Note, that the Penalty Payment is not automatic, the supervisor requests the penalty in the case the recipient refuses to comply without good reason
- **For an instance like Aquila, the fine can be expected to be significant**
  - Note, that due to how responsibility works in Finnish legislation, it is unlikely Visma products are sanctioned
- This is still good to keep in mind, especially when making contracts or if a product is determined to fall under legislation

Additional content

Lack of basic Accessibility and Usability increase Bounce rate

- **Bounce rate**, or the percentage of people landing on one page of your website and leave without visiting another, is a rough metric that measures if a user considers **"being in a relevant place"**
- Bounce rate as a measurement is in no way accurate nor does it apply to all products or services,
- However, in applicable cases, **a high bounce rate in the 70-80% range tell that there is something wrong with the website,** often the cause is in the complexity of the website or SEO
- IF a website for a product or service is non-informative, complicated or **difficult to access**, the user is simply incentivized to move on, increasing bounce rate

What is Bounce Rate?
https://miro.com/app/board/uXjVKV2HBeo=/?moveToWidget=3458764603985498276&cot=14

Why accessibility is important from a business value perspective
What is Bounce Rate?
https://miro.com/app/board/uXjVKV2HBeo=/?moveToWidget=3458764603985498276&cot=14

Enhanced UX Accessibility to Recapture 90% of Bounced Visitors
What is Bounce Rate?
https://miro.com/app/board/uXjVKV2HBeo=/?moveToWidget=3458764603985498276&cot=14

6 ways digital accessibility is good for business
https://adaptagency.com/why-accessibility-is-important-from-a-business-value-perspective

Laki digitaalisten palvelujen tarjoamisesta (306/2019)
https://www.finlex.fi/fi/laki/alkup/2019/20190306#Pidm46111192111744

European Accessibility Act
https://employment-social-affairs.ec.europa.eu/policies-and-activities/social-protection-social-inclusion/persons-disabilities/union-equality-strategy-rights-persons-disabilities-2021-2030/european-accessibility-act_en

User challenges

- Deficiencies in basic Accessibility link directly to **loss of user retention** and more importantly **disables certain users from using a product completely**
    - Screen reader users and keyboard users in particular
    - Most features that enable these users on a site are relatively simple and cost effective, as long as the necessary steps are embedded into the product early
- **Complicated user interfaces disincentivizes users from using your website**,
    - Users simply do not want to use a difficult or complicated websites
    - Due to internet, there is likely to be a more pleasant experience available
- The high speed nature of modern way of life drives people to use fast, uncomplicated services
    - The decision to buy or use a website is less likely to happen on complicated websites

Cost of Accessibility

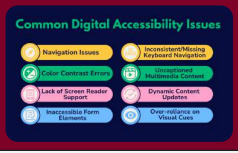
- The central question with accessibility is often **the cost** involved with accessibility
- When accessibility is implemented into products from the beginning the cost is negligible
    - Mainly a question of good work methodology and good design principles
- Implementing accessibility afterwards is however **always expensive and complicated**
    - Legacy code, unexpected interconnections, product bloat
    - Rewrites, redesigns, refactoring
- Bad accessibility cause **loss of revenue, as the customers simply refuse to buy**
    - Customer responsible for the accessibility in their product
    - Public sector companies and entities require accessibility standards to be fulfilled
- Sanctions, while unlikely to be placed on our products directly, will affect **our customers instead**
    - The effect on us would be equally devastating
    - Customer **very likely to change product**
    - If our product is mentioned to be the cause of the sanctions, the public relations fallout is likely to be significant

Law and Companies

- Most accessibility legislation is based on the harmonized EU directive, EN 301 549, therefore the national legislation might differ between different member states
    - For instance, the basis and size of **enforcement and penalties** might vary significantly between member nations
    - Thus a single accessibility issue **can be penalized several times, once in every country**
    - This also means that the responsibility of accessibility might differ in other countries from the Finnish version
- In Finland, the law is **Digital Services Act,** which states that the sanction falls under **Penalty Payment Act (Uhkasakkolaki)**
- If a sanction is placed on a product, the fine is calculated with the **capabilities of the product's company, not the product.**
    - Note, that the Penalty Payment is not automatic, the supervisor requests the penalty in the case the recipient refuses to comply without good reason
- **For an instance like Aquila, the fine can be expected to be significant**
    - Note, that due to how responsibility works in Finnish legislation, it is unlikely Visma products are sanctioned
- This is still good to keep in mind, especially when making contracts or if a product is determined to fall under legislation

User challenges

- Deficiencies in basic Accessibility link directly to **loss of user retention** and more importantly **disables certain users from using a product completely**
  - Screen reader users and keyboard users in particular
  - Most features that enable these users on a site are relatively simple and cost effective, as long as the necessary steps are embedded into the product early
- **Complicated user interfaces disincentivizes users from using your website**;
  - Users simply do not want to use a difficult or complicated websites
  - Due to internet, there is likely to be a more pleasant experience available
- The high speed nature of modern way of life drives people to use fast, uncomplicated services
  - The decision to buy or use a website is less likely to happen on complicated websites



Cost of Accessibility

- The central question with accessibility is often **the cost** involved with accessibility
- When accessibility is implemented into products from the beginning the cost is negligible
  - Mainly a question of good work methodology and good design principles
- Implementing accessibility afterwards is however **always expensive and complicated**
  - Legacy code, unexpected interconnections, product bloat
  - Rewrites, redesigns, refactoring
- Bad accessibility cause **loss of revenue, as the customers simply refuse to buy**
  - Customer responsible for the accessibility in their product
  - Public sector companies and entities require accessibility standards to be fulfilled
- Sanctions, while unlikely to be placed on our products directly, will affect **our customers instead**
  - The effect on us would be equally devastating
  - Customer **very likely to change product**
  - If our product is mentioned to be the cause of the sanctions, the public relations fallout is likely to be significant

Law and Companies

- Most accessibility legislation is based on the harmonized EU directive, EN 301 549, therefore the national legislation might differ between different member states
  - For instance, the basis and size of **enforcement and penalties** might vary significantly between member nations
  - Thus a single accessibility issue **can be penalized several times, once in every country**
  - This also means that the responsibility of accessibility might differ in other countries from the Finnish version
- In Finland, the law is **Digital Services Act,** which states that the sanction falls under **Penalty Payment Act (Uhkasakkolaki)**
- If a sanction is placed on a product, the fine is calculated with the **capabilities of the product's company, not the product.**
  - Note, that the Penalty Payment is not automatic, the supervisor requests the penalty in the case the recipient refuses to comply without good reason
- **For an instance like Aquila, the fine can be expected to be significant**
  - Note, that due to how responsibility works in Finnish legislation, it is unlikely Visma products are sanctioned
- This is still good to keep in mind, especially when making contracts or if a product is determined to fall under legislation

Additional content

Lack of basic Accessibility and Usability increase Bounce rate

- **Bounce rate**, or the percentage of people landing on one page of your website and leave without visiting another, is a rough metric that measures if a user considers **"being in a relevant place"**
- Bounce rate as a measurement is in no way accurate nor does it apply to all products or services,
- However, in applicable cases, **a high bounce rate in the 70-80% range tell that there is something wrong with the website**, often the cause is in the complexity of the website or SEO
- IF a website for a product or service is non-informative, complicated or **difficult to access**, the user is simply incentivized to move on, increasing bounce rate

What is Bounce Rate?
https://miro.com/app/board/uXjVKV2HBeo=/?moveToWidget=3458764603985498276&cot=14

Why accessibility is important from a business value perspective
What is Bounce Rate?
https://miro.com/app/board/uXjVKV2HBeo=/?moveToWidget=3458764603985498276&cot=14

Enhanced UX Accessibility to Recapture 90% of Bounced Visitors
What is Bounce Rate?
https://miro.com/app/board/uXjVKV2HBeo=/?moveToWidget=3458764603985498276&cot=14

6 ways digital accessibility is good for business
https://adaptagency.com/why-accessibility-is-important-from-a-business-value-perspective

Laki digitaalisten palvelujen tarjoamisesta (306/2019)
https://www.finlex.fi/fi/laki/alkup/2019/20190306#Pidm46111192111744

European Accessibility Act
https://employment-social-affairs.ec.europa.eu/policies-and-activities/social-protection-social-inclusion/persons-disabilities/union-equality-strategy-rights-persons-disabilities-2021-2030/european-accessibility-act_en

Tauko

10:34 - 10:44

How to do accessibility?

A clear goal for the acceptance level: weak vs. required by law vs. good.

- You should never accept your product to stay in any level of weak. Basic accessibility functions will have to be implemented in most products and services eventually. Striving for even better is of course always recommended.

Division of responsibility, accessibility requires cooperation from a variety of roles.

- No single person can ensure accessibility in a product or service (at least in most cases), it is a team effort that requires design, implementation, management and testing among its primary contributors.

More on responsibilities: https://advancedbyte.com/who-is-responsible-for-accessibility/

Coordination, e.g. Accessibility team vs. Coordinator.

- Adding to the previous point, not only do the different aspects of practical accessibility require accountable persons, the overall process of it does too.

Process consideration: assess the accessibility status, address issues, design and implement, test, perform monitoring, upkeep, and support.

- While there can be a multitude of steps taken in different parts of the accessibility process, the main phases remain the same. If you're only starting to bring accessibility to an existing product/service, there will be additional steps like determining the initial level of a11y, conducting training and establishing roles and processes. Once you include accessibility work into your work process, the routine looks something like this.

https://medium.com/publicissapient/accessibility-in-software-development-ef3a26c1d0b7

Accessibility must be considered in all phases of product development

- Just as everyone on the team should be aware of accessibility, so should it influence their work. Sometimes to a smaller, sometimes to a larger degree.

https://devblogs.microsoft.com/premier-developer/accessibility-an-all-stages-a-good-for-all-kitchen-approach/

Accessibility does not need to be a massive "extra" effort.

- Designers and front-end developers are primarily responsible for creating accessibility in practice. Fortunately, best practices for both tend to include accessibility considerations. More often than not, if accessibility issues surface, it's due to old implementations, bad implementations, bad design or other forms of human laziness/errors instead of any other reason.

Image source: https://www.geeksforgeeks.org/difference-between-front-end-developer-and-ui-ux-designer/

If you do things properly and are willing to fix the things that aren't done so, accessibility laws and authorities will have few sanctioning opportunities.

- Especially with older software and services there can be quite a laundry list of issues, but clear communication and an effort to fix them will usually be enough.

Image source: https://www.interactivesoftware.com/blog/accessible-communication/

To ensure your product/service is accessible, you need to follow WCAG guidelines.

- WCAG (or Web Content Accessibility Guidelines) have a few different levels, A/AA/AAA, of which the double A is the goal for most products. In-depth dive into this in chapter 8.

Image source: https://www.audioeye.com/post/what-new-with-wcag-2-2/

Every product is required to have an accessibility statement, a feedback channel and instructions on reporting to the national supervisor.

Accessibility is not a single shot effort. Whether you like it or not, it has to be a constant part of your development process.

Additional content

Accessibility statement
- Description of product accessibility level
- Listed deficiencies must be addressed and they may not be endlessly postponed
- Feedback channel for customers and users to give accessibility related feedback, which must be answered within 14 days (e.g. "Thank you for your feedback" is often enough)
-Who made the assessment, including if it was an internal or external auditor
- Instructions on how to do a request for investigation or a complaint with AVI
- Must be upgraded yearly

Exceptions

End-of-life-products do not fall under the accessibility requirements sphere. However, the EoL status of a product must be clearly stated as well as what accessibility problems are present and that they won't be addressed.

Good hands-on tips and hints : https://www.interaction-design.org/literature/topics/accessibility

A clear goal for the acceptance level: weak vs. required by law vs. good.

- You should never accept your product to stay in any level of weak. Basic accessibility functions will have to be implemented in most products and services eventually. Striving for even better is of course always recommended.

Division of responsibility, accessibility requires cooperation from a variety of roles.

- No single person can ensure accessibility in a product or service (at least in most cases), it is a team effort that requires design, implementation, management and testing among its primary contributors.

More on responsibilities: https://advancedbytez.com/who-is-responsible-for-accessibility/

Coordination, e.g. Accessibility team vs. Coordinator.

- Adding to the previous point, not only do the different aspects of practical accessibility require accountable persons, the overall process of it does too.

Process consideration: assess the accessibility status, address issues, design and implement, test, perform monitoring, upkeep, and support.

- While there can be a multitude of steps taken in different parts of the accessibility process, the main phases remain the same. If you're only starting to bring accessibility to an existing product/service, there will be additional steps like determining the initial level of a11y, conducting training and establishing roles and processes. Once you include accessibility work into your work process, the routine looks something like this.

https://medium.com/@talktoresh/accessibility-in-software-development-e73acbc648b7

Accessibility must be considered in all phases of product development

- Just as everyone on the team should be aware of accessibility, so should it influence their work. Sometimes to a smaller, sometimes to a larger degree.

https://devblogs.microsoft.com/premier-developer/accessibility-at-all-stages-a-good-for-all-sdlc-approach/

Accessibility does not need to be a massive "extra" effort.

- Designers and front-end developers are primarily responsible for creating accessibility in practice. Fortunately, best practices for both tend to include accessibility considerations. More often than not, if accessibility issues surface, it's due to old implementations, bad implementations, bad design or other forms of human laziness/errors instead of any other reason.

Image source: https://www.geeksforgeeks.org/difference-between-front-end-developer-and-ui-ux-designer/

If you do things properly and are willing to fix the things that aren't done so, accessibility laws and authorities will have few sanctioning opportunities.

- Especially with older software and services there can be quite a laundry list of issues, but clear communication and an effort to fix them will usually be enough.

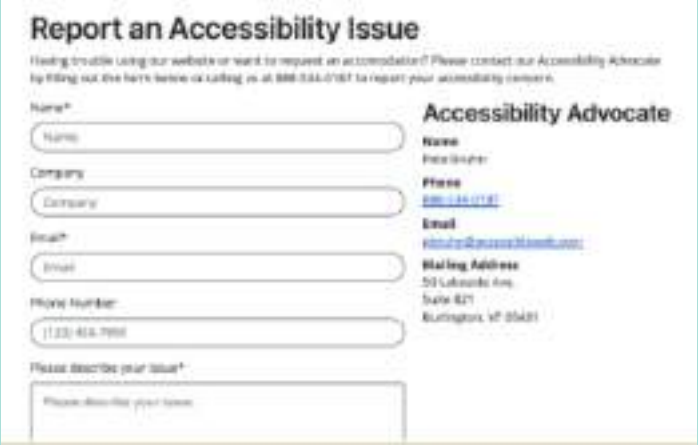
Image source: https://www.interactsoftware.com/blog/accessible-communication/

To ensure your product/service is accessible, you need to follow WCAG guidelines.

- WCAG (or Web Content Accessibility Guidelines) have a few different levels, A/AA/AAA, of which the double A is the goal for most products. In-depth dive into this in chapter 8.

Image source: https://www.audioeye.com/post/whats-new-with-wcag-2-2/

Every product is required to have an accessibility statement, a feedback channel and instructions on reporting to the national supervisor.



Here's how to contact the relevant authorities for reporting accessibility misconduct:

1. Lorem ipsum
2. Dolor et amet
3. ...
4. ...

Accessibility is not a single shot effort. Whether you like it or not, it has to be a constant part of your development process.

How to do accessibility?

A clear goal for the acceptance level: weak vs. required by law vs. good.

- You should never accept your product to stay in any level of weak. Basic accessibility functions will have to be implemented in most products and services eventually. Striving for even better is of course always recommended.

Division of responsibility, accessibility requires cooperation from a variety of roles.

- No single person can ensure accessibility in a product or service (at least in most cases), it is a team effort that requires design, implementation, management and testing among its primary contributors.

More on responsibilities: https://advancedbyez.com/who-is-responsible-for-accessibility/

Coordination, e.g. Accessibility team vs. Coordinator.

- Adding to the previous point, not only do the different aspects of practical accessibility require accountable persons, the overall process of it does too.

Process consideration: assess the accessibility status, address issues, design and implement, test, perform monitoring, upkeep, and support.

- While there can be a multitude of steps taken in different parts of the accessibility process, the main phases remain the same. If you're only starting to bring accessibility to an existing product/service, there will be additional steps like determining the initial level of a11y, conducting training and establishing roles and processes. Once you include accessibility work into your work process, the routine looks something like this.

https://medium.com/@adbenchdev/accessibility-in-software-development-a7bc0c6b17

Accessibility must be considered in all phases of product development

- Just as everyone on the team should be aware of accessibility, so should it influence their work. Sometimes to a smaller, sometimes to a larger degree.

https://devblogs.microsoft.com/premier-developer/accessibility-in-all-stages-a-good-for-all-sdlc-approaches/

Accessibility does not need to be a massive "extra" effort.

- Designers and front-end developers are primarily responsible for creating accessibility in practice. Fortunately, best practices for both tend to include accessibility considerations. More often than not, if accessibility issues surface, it's due to old implementations, bad implementations, bad design or other forms of human laziness/errors instead of any other reason.

Image source: https://www.geeksforgeeks.org/difference-between-front-end-developer-and-ui-ux-designer/

If you do things properly and are willing to fix the things that aren't done so, accessibility laws and authorities will have few sanctioning opportunities.

- Especially with older software and services there can be quite a laundry list of issues, but clear communication and an effort to fix them will usually be enough.

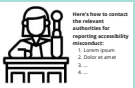
Image source: https://www.interactsoftware.com/blog/accessible-communication/

To ensure your product/service is accessible, you need to follow WCAG guidelines.

- WCAG (or Web Content Accessibility Guidelines) have a few different levels, A/AA/AAA, of which the double A is the goal for most products. In-depth dive into this in chapter 8.

Image source: https://www.audioeye.com/post/whats-new-with-wcag-2-2/

Every product is required to have an accessibility statement, a feedback channel and instructions on reporting to the national supervisor.



Here's how to contact the relevant authorities for reporting accessibility misconduct:
1. Lorem ipsum
2. Dolor er amet
3. ...
4. ...

Accessibility is not a single shot effort. Whether you like it or not, it has to be a constant part of your development process.

Additional content

Accessibility statement
- Description of product accessibility level
- Listed deficiencies must be addressed and they may not be endlessly postponed
- Feedback channel for customers and users to give accessibility related feedback, which must be answered within 14 days (e.g. "Thank you for your feedback" is often enough)
-Who made the assessment, including if it was an internal or external auditor
- Instructions on how to do a request for investigation or a complaint with AVI
- Must be upgraded yearly

Exceptions

End-of-life-products do not fall under the accessibility requirements sphere. However, the EoL status of a product must be clearly stated as well as what accessibility problems are present and that they won't be addressed.

Good hands-on tips and hints : https://www.interaction-design.org/literature/topics/accessibility

Web Content Accessibility Guidelines, or WCAG

The Web Content Accessibility Guidelines (WCAG) are technical standards on web accessibility developed by the World Wide Web Consortium (W3C). The guidelines represent a shared, international standard developed **BY THEIR INTENDED** stakeholders, including **VARIOUS DISABILITY** organizations, government, and accessibility research organizations. WCAG isn't a legal requirement, but a set of globally adopted standards that provide a common framework for making digital content and online digital assets accessible to people with a broad range of disabilities, including sensory, intellectual, learning and physical disabilities.

Introduction to WCAG: https://www.w3.org/WAI/standards-guidelines/wcag/

Past, present & future

WCAG 1.0 – Born in May 1999 and included 14 categories as well as the A, AA and AAA conformance level hierarchy.

WCAG 2.0 – Published in December 2008, introduced the principles of perceivable, operable, understandable and robust and redefined the A, AA and AAA levels of conformance.

WCAG 2.1 – Published in June 2018, it amended or added criteria to address accessibility concerns for people with cognitive, language, learning, low vision and other disabilities, as well as for people using mobile devices, on tablets and introduced 17 new success criteria.

Currently in use

WCAG 2.2 – Released in October 2023 and added categories to cover topics such as mobile use, cognitive disabilities and authentication with increasing one or two attribute categories such as parsing.

WCAG 3.0 – Still under construction and is estimated to be ready – eventually (like years – maybe). It aims to introduce a rigorous 2.0 rather, but rather the two will coexist for some time. WCAG 3 will have similar fundamental and specific accessibility principles as WCAG 2.0 but will have a very different structure, different conformance model, and broader scope.

The following success criteria are new in WCAG 2.2:

- 2.4.11 Focus Not Obscured (Minimum) (AA)
- 2.4.12 Focus Not Obscured (Enhanced) (AAA)
- 2.4.13 Focus Appearance (AAA)
- 3.2.6 Consistent Help (A)
- 3.3.7 Redundant Entry (A)
- 3.3.8 Accessible Authentication (Minimum) (AA)
- 3.3.9 Accessible Authentication (Enhanced) (AAA)
- 2.5.7 Dragging Movements (AA)
- 2.5.8 Target Size (Minimum) (AA)

WCAG 2.2 builds on and is backwards compatible with WCAG 2.1, meaning web pages that conform to WCAG 2.2 are at least as accessible as pages that conform to WCAG 2.1. Requirements have been added that focus on 1.1 and 2.5. WCAG 2.2 has removed one success criterion, 4.1.1 Parsing.

WCAG essentials

The 4 principles of accessibility

The guidelines and success criteria are organised around the following four principles, with every the foundation necessary for anyone to access with use their content. Content is perceivable to use, therefore it and easily understandable.

Perceivable
Information and user interface components must be presented to users in ways they can perceive. It means that users must be able to perceive the information being presented (not to be invisible to all their senses).

Operable
User interface components and navigation must be operable. This means that users must be able to operate the interface (for example, the functionality must be available from a keyboard).

Understandable
Information and the operation of user interface must be understandable. This means users must be able to understand the information as well as the operation of the user interface (the content or operation cannot be beyond their understanding).

Robust
Content must be robust enough that it can be interpreted reliably by a wide variety of users, including assistive technologies – and remain accessible as technologies advance. The more features browsers and other user agents support, the more assistive technologies access the information they need.

Users' ability to recognise and use the information on the screen available to them.

Users' ability to find and utilise all the functionality and tasks of a site.

Users' ability to clearly understand the information on the site and service features.

Users' ability to reliably interpret the content regardless of the ever-evolving set of means of access.

If any of these are not true, users with disabilities will not be able to use the Web.

New versions of WCAG continue to build on previous versions. This means that future WCAG version numbers will mark a set of requirements that WCAG 2.1 and 2.2 which, in turn, inherit requirements from previous WCAG 2 versions. Requirements structure the overall framework of guidelines and ensure backwards compatibility.

Understanding WCAG: https://www.w3.org/WAI/WCAG22/Understanding/

WCAG layers of guidance

The checkpoints and sub-guidelines are not arbitrary or widely and include field designers and developers, policy makers, purchasing agents, authors, and educators. In order to meet the varying needs of the audience, a wide-angle lens approach is provided (including: based principles, general guidelines, testable success criteria and a rich collection of sufficient techniques, advisory techniques, and common failures) which will provide the flexibility to be practical.

All of the layers of guidance (principles, guidelines, success criteria, and sufficient and advisory techniques) are applicable to all Web pages and sites. Authors can choose to meet success criteria by using the techniques, or can use other techniques that satisfy the success criteria to the extent that they are able and wish to do so.

WCAG layers of guidance

Principles
The principles are the ones that provide the foundation for Web accessibility (perceivable, operable, understandable, and robust). See also: https://www.w3.org/TR/wcag-3.0/#principles

WCAG layers of guidance

Guidelines
Guidelines are principles, or guidelines. They (guidelines) provide the basic goals that a person, conscientious research or not, can take, between levels accessible to users with different disabilities. The guidelines are not testable, but provide the framework and overall objectives to help authors understand the success criteria and better implement the techniques.

WCAG layers of guidance

Success Criteria
Success criteria are testable, success criteria are prerequisites or other limits, or to be used when a requirement or a conformance succeeding any necessary such as or design specifications, purchasing, regulations, and contractual requirements. Success criteria are testable, but not to be so specific that they are not applicable to all Web pages and sites. Success criteria are technology-neutral, and are intended to be testable with or without the use of assistive technology, and, where technologies, authors can choose to use the techniques.

WCAG layers of guidance

Sufficient and advisory techniques
As part of the guidelines and success criteria in the Web site, it (the content) must, the working group has also developed a wide variety of techniques. The techniques are informative and that not one category. These that are sufficient for meeting the success criteria, and those that are advisory. Sufficient techniques are those that are known to be sufficient for meeting the success criteria and advisory or better, advisory. The guidelines, success criteria, and sufficient techniques are testable but not so specific that they are not applicable to all Web pages and sites. Advisory techniques are not testable and are intended to go beyond the success criteria, or address barriers not addressed by success criteria.

Sufficient and advisory techniques: https://www.w3.org/WAI/WCAG22/Techniques/

Common failures
Common failures are failures, or failure conditions, which are known to be a common or frequent mistake or error, or are likely to cause an error or design specifications, purchasing, regulations, and contractual requirements. In order to meet the needs of different groups, or of different circumstances, these lists of conformance requirements are known. Common failures are informative.

Common failures: https://www.w3.org/WAI/WCAG22/Techniques/failures/Overview.html

Accessibility
Accessibility principles are guidelines. The (guidelines) provide the basic goals that a person, conscientious research or not, can take, between levels accessible to users with different disabilities. The guidelines are not testable, but provide the framework and overall objectives to help authors understand the success criteria and better implement the techniques.

Principles
Principles are the ones that provide the foundation for Web accessibility (perceivable, operable, understandable, and robust). See also: https://www.w3.org/TR/wcag-3.0/#principles

Note that even conformance criteria are the highest level, which will not be accessible to individuals with all types, degrees, or combinations of disabilities, particularly in the cognitive, language and learning groups. Authors who wish to go beyond the minimum levels of accessibility, including the advisory techniques, should consult the Web Content Accessibility Guidelines (WCAG) 2 Quick Reference, which is a filtered version of the Web Content Accessibility Guidelines, as it is a quick reference and lists all techniques, including the advisory techniques, and provides a quick way to see which techniques are available for each topic.

WCAG layers of guidance

Sufficient and advisory techniques
As part of the guidelines and success criteria in the Web site, it (the content) must, the working group has also developed a wide variety of techniques. The techniques are informative and that not one category. These that are sufficient for meeting the success criteria, and those that are advisory. Sufficient techniques are those that are known to be sufficient for meeting the success criteria and advisory or better, advisory. The guidelines, success criteria, and sufficient techniques are testable but not so specific that they are not applicable to all Web pages and sites. Advisory techniques are not testable and are intended to go beyond the success criteria, or address barriers not addressed by success criteria.

Sufficient and advisory techniques: https://www.w3.org/WAI/WCAG22/Techniques/

WCAG layers of guidance

Sufficient and advisory techniques
As part of the guidelines and success criteria in the Web site, it (the content) must, the working group has also developed a wide variety of techniques. The techniques are informative and that not one category. These that are sufficient for meeting the success criteria, and those that are advisory. Sufficient techniques are those that are known to be sufficient for meeting the success criteria and advisory or better, advisory. The guidelines, success criteria, and sufficient techniques are testable but not so specific that they are not applicable to all Web pages and sites. Advisory techniques are not testable and are intended to go beyond the success criteria, or address barriers not addressed by success criteria.

Sufficient and advisory techniques: https://www.w3.org/WAI/WCAG22/Techniques/

Common failures
Common failures are failures, or failure conditions, which are known to be a common or frequent mistake or error, or are likely to cause an error or design specifications, purchasing, regulations, and contractual requirements. In order to meet the needs of different groups, or of different circumstances, these lists of conformance requirements are known. Common failures are informative.

Common failures: https://www.w3.org/WAI/WCAG22/Techniques/failures/Overview.html

Accessibility
Accessibility principles are guidelines. The (guidelines) provide the basic goals that a person, conscientious research or not, can take, between levels accessible to users with different disabilities. The guidelines are not testable, but provide the framework and overall objectives to help authors understand the success criteria and better implement the techniques.

Principles
Principles are the ones that provide the foundation for Web accessibility (perceivable, operable, understandable, and robust). See also: https://www.w3.org/TR/wcag-3.0/#principles

Note that even conformance criteria are the highest level, which will not be accessible to individuals with all types, degrees, or combinations of disabilities, particularly in the cognitive, language and learning groups. Authors who wish to go beyond the minimum levels of accessibility, including the advisory techniques, should consult the Web Content Accessibility Guidelines (WCAG) 2 Quick Reference, which is a filtered version of the Web Content Accessibility Guidelines, as it is a quick reference and lists all techniques, including the advisory techniques, and provides a quick way to see which techniques are available for each topic.

All of our products should aim for the AA-standard of WCAG in terms of accessibility. This means fulfilling all of the criteria for both A- and AA-levels.

WCAG Success Criteria

- **Level A** – Includes the most essential (and least strict) requirements.
- **Level AA** – Includes additional criteria. Most websites should aim for Level AA conformance.
- **Level AAA** – Includes the most strict requirements. Some types of content may be unable to meet all Level AAA success criteria.

Image sources: https://www.w3.org/TR/wcag-3.0/#principles

© 2022 Codegarden Oy

Web Content Accessibility Guidelines, or WCAG

The Web Content Accessibility Guidelines (WCAG) are technical standards on web accessibility developed by the World Wide Web Consortium (W3C). The guidelines represent a shared, international standard developed by many different stakeholders, including industry, disability organizations, government, and accessibility research organizations. WCAG isn't a legal requirement, but a set of globally adopted standards. The guidelines aim to make websites, apps, electronic documents, and other digital assets accessible to people with a broad range of disabilities, including sensory, intellectual, learning and physical disabilities.

Introduction to WCAG: https://www.w3.org/WAI/standards-guidelines/wcag/

Past, present & future

WCAG 1.0

Born in May 1999 and included 14 categories as well as the A, AA and AAA conformance level hierarchy.

WCAG 2.0

Published in December 2008, introduced the principles of perceivable, operable, understandable and robust and redefined the A, AA and AAA levels of conformance.

WCAG 2.1

Published in June 2018, is considered an interim measure to provide updated standards reflecting advancements in digital content since the publication of 2.0. The primary differences between WCAG 2.0 and WCAG 2.1 concern mobile devices or tablets and introduced 17 new success criteria.

WCAG 2.2

Released in October 2023 and added categories to cover topics such as mobile use, focus and accessible authentication while removing one or two obsolete categories such as parsing.

Currently in use

WCAG 3.0

Still under construction and is estimated to be ready... eventually (few years maybe). It won't immediately replace 2.2 either, but rather the two will coexist for some time. WCAG 3 will have similar fundamental and specific accessibility requirements as WCAG 2. WCAG 3 will have a different structure, different conformance model, and broader scope.

The following success criteria are new in WCAG 2.2:

- 2.4.11 Focus Not Obscured (Minimum) (AA)
- 2.4.12 Focus Not Obscured (Enhanced) (AAA)
- 2.4.13 Focus Appearance (AAA)
- 2.5.7 Dragging Movements (AA)
- 2.5.8 Target Size (Minimum) (AA)
- 3.2.6 Consistent Help (A)
- 3.3.7 Redundant Entry (A)
- 3.3.8 Accessible Authentication (Minimum) (AA)
- 3.3.9 Accessible Authentication (Enhanced) (AAA)

WCAG 2.2 builds on and is backwards compatible with WCAG 2.1, meaning web pages that conform to WCAG 2.2 are at least as accessible as pages that conform to WCAG 2.1. Requirements have been added that build on 2.1 and 2.0. WCAG 2.2 has removed one success criterion, 4.1.1 Parsing.

WCAG essentials

The 4 principles of accessibility

The guidelines and success criteria are organized around the following four principles, which lay the foundation necessary for anyone to access and use Web content. Anyone who wants to use the Web must have content that is:

Perceivable
Information and user interface components must be presentable to users in ways they can perceive. This means that users must be able to perceive the information being presented (it can't be invisible to all of their senses)

Operable
User interface components and navigation must be operable. This means that users must be able to operate the interface (the interface cannot require interaction that a user cannot perform)

Understandable
Information and the operation of user interface must be understandable. This means that users must be able to understand the information as well as the operation of the user interface (the content or operation cannot be beyond their understanding)

Robust
Content must be robust enough that it can be interpreted reliably by a wide variety of user agents, including assistive technologies. This means that users must be able to access the content as technologies advance (as technologies and user agents evolve, the content should remain accessible)

https://www.developer.tech.gov.sg/guidelines/standards-and-best-practices/wcag-2-1.html

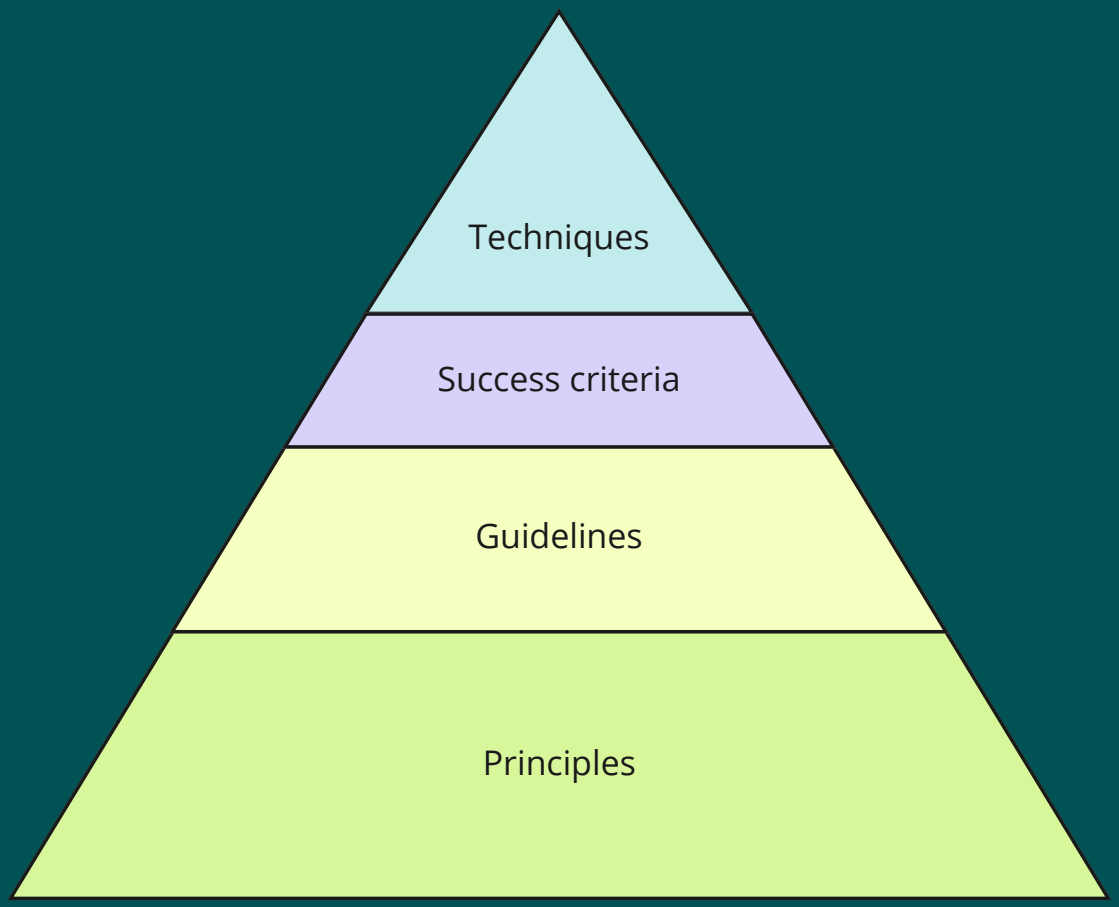
If any of these are not true, users with disabilities will not be able to use the Web.

New versions of WCAG continue to build on previous versions. This means that future WCAG version numbers will meet a set of requirements for WCAG 2.2 which, in turn, inherit requirements from previous WCAG 2 versions. Requirements structure the overall framework of guidelines and ensure backwards compatibility.

Understanding WCAG: https://www.w3.org/WAI/WCAG22/Understanding/intro#understanding-the-four-principles-of-accessibility

WCAG layers of guidance

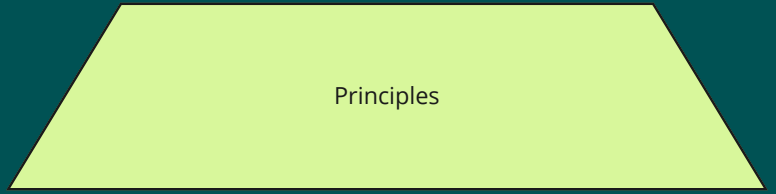
The individuals and organizations that use WCAG vary widely and include Web designers and developers, policy makers, purchasing agents, teachers, and students. In order to meet the varying needs of this audience, several layers of guidance are provided including overall *principles*, general *guidelines*, testable *success criteria* and a rich collection of *sufficient techniques*, *advisory techniques*, and *documented common failures* with examples, resource links and code.



All of these layers of guidance (principles, guidelines, success criteria, and sufficient and advisory techniques) work together to provide guidance on how to make content more accessible. Authors are encouraged to view and apply all layers that they are able to, including the advisory techniques, in order to best address the needs of the widest possible range of users.

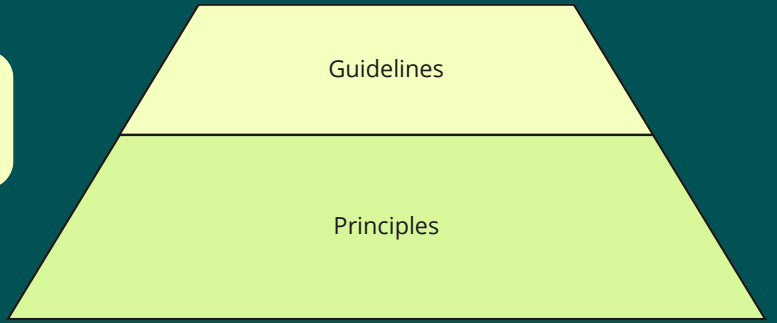
WCAG layers of guidance

Principles
At the core are four principles that provide the foundation for Web accessibility: *perceivable, operable, understandable, and robust*. See also Understanding the Four Principles of Accessibility.



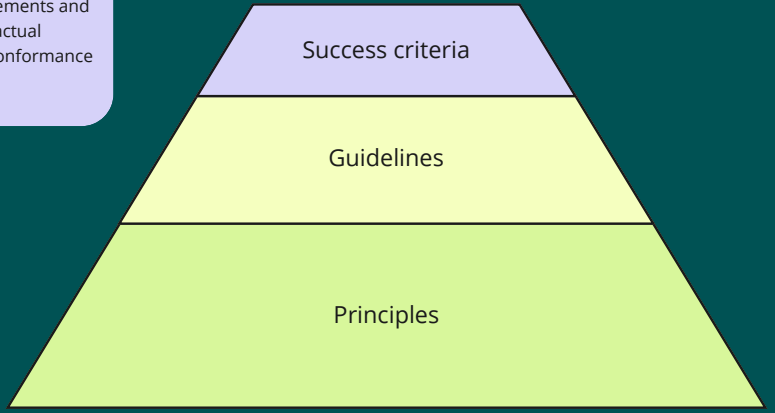
WCAG layers of guidance

Guidelines
Building on the principles are guidelines. The 13 guidelines provide the basic goals that authors should work toward in order to make content more accessible to users with different disabilities. The guidelines are not testable, but provide the framework and overall objectives to help authors understand the success criteria and better implement the techniques.



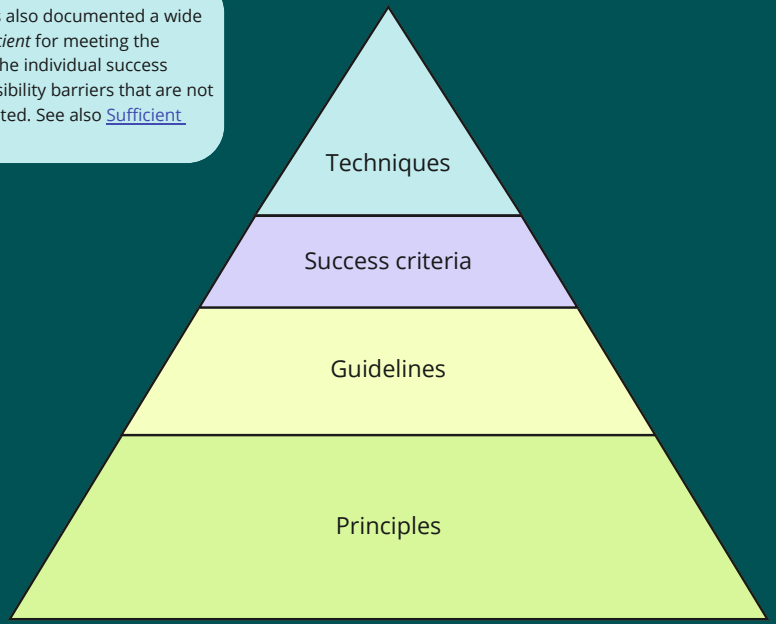
WCAG layers of guidance

Success Criteria
For each guideline, testable success criteria are provided to allow WCAG 2.2 to be used where requirements and conformance testing are necessary such as in design specification, purchasing, regulation, and contractual agreements. In order to meet the needs of different groups and different situations, three levels of conformance are defined: A (lowest), AA, and AAA (highest). Additional information on WCAG levels can be found in Understanding Levels of Conformance.



WCAG layers of guidance

Sufficient and Advisory Techniques
For each of the *guidelines* and *success criteria* in the WCAG 2.2 document itself, the working group has also documented a wide variety of *techniques*. The techniques are informative and fall into two categories: those that are *sufficient* for meeting the success criteria and those that are *advisory*. The advisory techniques go beyond what is required by the individual success criteria and allow authors to better address the guidelines. Some advisory techniques address accessibility barriers that are not covered by the testable success criteria. Where common failures are known, these are also documented. See also Sufficient and Advisory Techniques in Understanding WCAG 2.2.



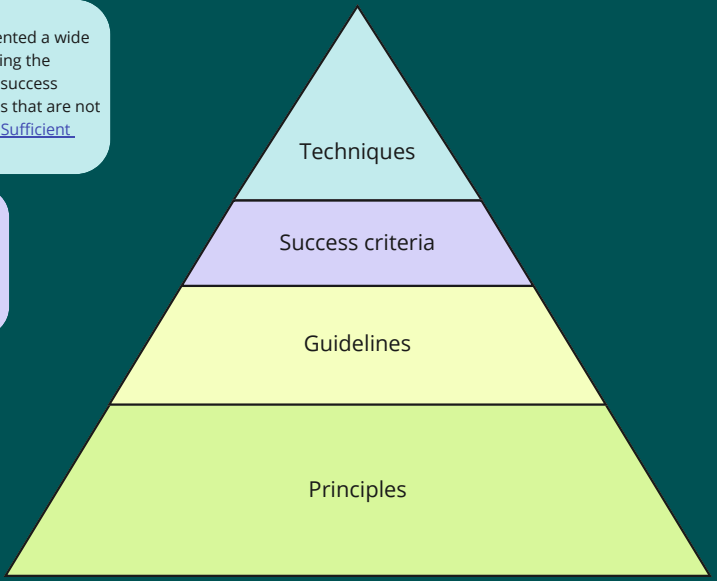
WCAG layers of guidance

Sufficient and Advisory Techniques
For each of the *guidelines* and *success criteria* in the WCAG 2.2 document itself, the working group has also documented a wide variety of *techniques*. The techniques are informative and fall into two categories: those that are *sufficient* for meeting the success criteria and those that are *advisory*. The advisory techniques go beyond what is required by the individual success criteria and allow authors to better address the guidelines. Some advisory techniques address accessibility barriers that are not covered by the testable success criteria. Where common failures are known, these are also documented. See also Sufficient and Advisory Techniques in Understanding WCAG 2.2.

Success Criteria
For each guideline, testable success criteria are provided to allow WCAG 2.2 to be used where requirements and conformance testing are necessary such as in design specification, purchasing, regulation, and contractual agreements. In order to meet the needs of different groups and different situations, three levels of conformance are defined: A (lowest), AA, and AAA (highest). Additional information on WCAG levels can be found in Understanding Levels of Conformance.

Guidelines
Building on the principles are guidelines. The 13 guidelines provide the basic goals that authors should work toward in order to make content more accessible to users with different disabilities. The guidelines are not testable, but provide the framework and overall objectives to help authors understand the success criteria and better implement the techniques.

Principles
At the core are four principles that provide the foundation for Web accessibility: *perceivable, operable, understandable, and robust*. See also Understanding the Four Principles of Accessibility.



Note that even content that conforms at the highest level (AAA) will not be accessible to individuals with all types, degrees, or combinations of disability, particularly in the cognitive, language, and learning areas. Authors are encouraged to consider the full range of techniques, including the advisory techniques, Making Content Usable for People with Cognitive and Learning Disabilities, as well as to seek relevant advice about current best practice to ensure that Web content is accessible, as far as possible, to this community. Metadata may assist users in finding content most suitable for their needs.

All of our products should aim for the AA-standard of WCAG in terms of accessibility. This means fulfilling all of the criteria for both A- and AA-levels.

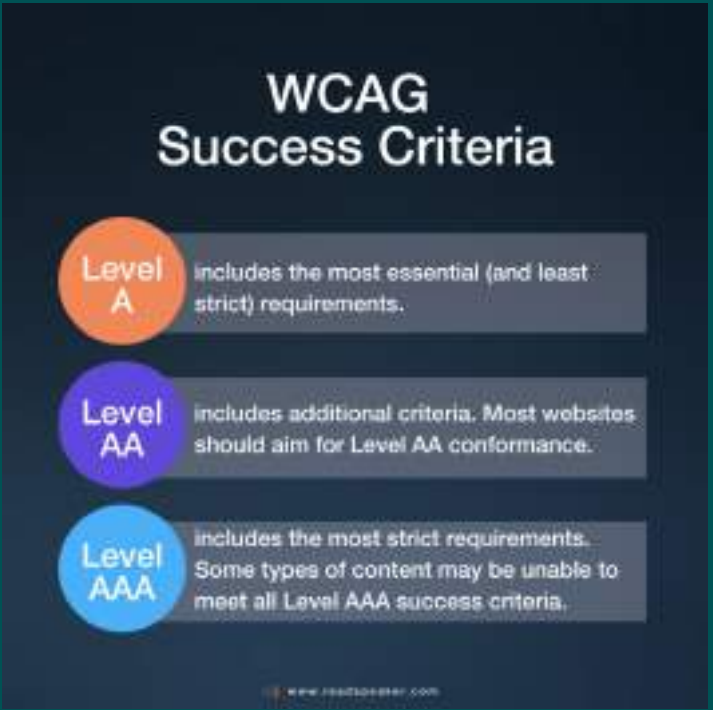


Image source: https://www.readspeaker.com/blog/web-accessibility-guide/

Web Content Accessibility Guidelines, or WCAG

The Web Content Accessibility Guidelines (WCAG) are technical standards on web accessibility developed by the World Wide Web Consortium (W3C). The guidelines represent a shared, international standard developed by many of the world's leading web accessibility experts, including representatives from the W3C Web Accessibility Initiative (WAI) and other organizations and individuals around the world. The guidelines are made websites, apps, electronic documents, and other digital assets accessible to people with a broad range of disabilities, including sensory, intellectual, learning and physical disabilities.

Introduction to WCAG https://www.w3.org/WAI/standards-guidelines/wcag/

Past, present & future

WCAG 1.0 Born in May 1999 and included 14 categories as well as the A, AA and AAA conformance level hierarchy.

WCAG 2.0 Published in December 2008, introduced the principles of perceivable, operable, understandable and robust, and redefined the A, AA and AAA levels of conformance.

WCAG 2.1 Published in June 2018, is considered an interim measure to provide updated standards reflecting advancements in digital content since the publication of 2.0. The primary differences between WCAG 2.0 and WCAG 2.1 concern mobile devices, low vision and dexterity. 17 new success criteria.

WCAG 2.2 Released in October 2020 and added categories to cover topics such as mobile use, focus and accessible authentication while removing one or two obsolete categories such as parsing.

Currently in use

WCAG 3.0 Still under construction and is extended-trole-ready, - eventually (they plan to) replace the previous versions of WCAG. Unlike WCAG 2.x, it is not backwards compatible. WCAG 3 will have more flexibility in accepting the overall conformance requirements as WCAG 2. WCAG 3 will have a different structure, different conformance model, and broader scope.

The following success criteria are new in WCAG 2.2:

- 2.4.11 (AA)
- 2.4.12 (AAA)
- 2.4.13 (AAA)
- 3.2.6 (A)
- 3.3.7 (A)
- 3.3.8 (AA)
- 3.3.9 (AAA)
- 2.5.7 (AA)
- 2.5.8 (AA)
- 3.2.5 (AAA)

WCAG 2.2 builds on and is backwards-compatible with WCAG 2.1, meaning web pages that conform to WCAG 2.1 are at least as accessible as pages that conform to WCAG 2.1. Requirements have been added that build on 2.1 and 2.0. WCAG 2.2 has removed one success criterion.

WCAG essentials

The 4 principles of accessibility

The guidelines and success criteria are organized around the following four principles, which lay the foundation necessary for a person to access or use content. In order to be a Web accessible site, the WCAG 2.0 Rules must make it that it:

perceivable – information and user interface components must be presented to users in ways they can perceive. This means that they must be able to perceive the information being presented (it can't be invisible to all of their senses).

operable – user interface components and navigation must be operable. This means that users must be able to operate the interface (the interface cannot require interaction that a user cannot perform).

understandable – information and the operation of user interface must be understandable. This means that users must be able to understand the information as well as the operation of the user interface (the content must be presented in a way that users can understand).

Robust – content must be robust enough that it can be interpreted reliably by a wide range of user agents, including assistive technologies. This means that users must be able to access the content as technologies advance (the content should still work when new technologies and user agents come out in the future).

If any of these are not true, users with disabilities will not be able to use the Web.

New versions of WCAG continue to build on previous versions. This means that future WCAG version numbers will meet a set of requirements for WCAG 2.x which, in turn, inherit requirements from previous WCAG 2 versions. Requirements structure the overall framework of guidelines and ensure backwards compatibility.

Understanding WCAG https://www.w3.org/WAI/WCAG21/Understanding/

WCAG layers of guidance

The individuals and organizations that use Web site and by extension, and those Web designers and developers, policy makers, purchasers, and others who are involved in the development process, have different needs and different levels of knowledge. Concepts building blocks include, general guidelines, success criteria, sample techniques and a rich collection of guidance techniques, advisory techniques, and technical techniques. Based from each principle, guidelines have specific success criteria.

All of these layers of guidance (principles, guidelines, success criteria, and sufficient and advisory techniques) work together to provide a framework for understanding how to make Web content more accessible to people with a wide range of disabilities. This includes including advisory techniques, in order to help address the needs of the wide and diverse range of users.

WCAG layers of guidance

Principles
At the very top are the principles or guidelines. The 4 guidelines provide the foundation for Web accessibility, perceivable, operable, understandable, and robust. Without these four principles, content is not accessible to people with disabilities. Principles are necessary but provide the framework and not enough on their own.

WCAG layers of guidance

Guidelines
Under the principles are guidelines. The 14 guidelines provide the foundation goals that authors should aim for in order to make content accessible to people with disabilities. The guidelines are necessary but provide the framework and not enough on their own.

Guidelines
Principles

WCAG layers of guidance

Success Criteria
Success criteria are testable success criteria, and provide a rule or criteria to be used when requirements and specifications are written. Success criteria are testable, and are written in such a way that they can be used where requirements and specifications are written. Success criteria are testable, and are written in such a way that they can be used where requirements and specifications are written. Success criteria are testable, and are written in such a way that they can be used where requirements and specifications are written.

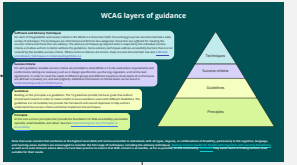
Success criteria
Guidelines
Principles

WCAG layers of guidance

Sufficient and advisory techniques
Under the success criteria are sufficient and advisory techniques. Sufficient techniques are a set of techniques that are sufficient for meeting the success criteria. Advisory techniques are other techniques that are not sufficient for meeting the success criteria, and allow authors to better address the guidelines. Some advisory techniques address accessibility barriers that are not covered by the success criteria and sufficient techniques. Advisory techniques address barriers that are not covered by the success criteria and sufficient techniques.

WCAG layers of guidance

Sufficient and advisory techniques
Under the success criteria are sufficient and advisory techniques. Sufficient techniques are a set of techniques that are sufficient for meeting the success criteria. Advisory techniques are other techniques that are not sufficient for meeting the success criteria, and allow authors to better address the guidelines. Some advisory techniques address accessibility barriers that are not covered by the success criteria and sufficient techniques. Advisory techniques address barriers that are not covered by the success criteria and sufficient techniques. Advisory techniques address barriers that are not covered by the success criteria and sufficient techniques.



Sufficient and advisory techniques
Under the success criteria are sufficient and advisory techniques. Sufficient techniques are a set of techniques that are sufficient for meeting the success criteria. Advisory techniques are other techniques that are not sufficient for meeting the success criteria, and allow authors to better address the guidelines. Some advisory techniques address accessibility barriers that are not covered by the success criteria and sufficient techniques. Advisory techniques address barriers that are not covered by the success criteria and sufficient techniques. Advisory techniques address barriers that are not covered by the success criteria and sufficient techniques.

Success Criteria
Success criteria are testable success criteria, and provide a rule or criteria to be used when requirements and specifications are written. Success criteria are testable, and are written in such a way that they can be used where requirements and specifications are written. Success criteria are testable, and are written in such a way that they can be used where requirements and specifications are written.

Guidelines
Under the principles are guidelines. The 14 guidelines provide the foundation goals that authors should aim for in order to make content accessible to people with disabilities. The guidelines are necessary but provide the framework and not enough on their own. Without these four principles, content is not accessible to people with disabilities. Principles are necessary but provide the framework and not enough on their own.

Principles
At the very top are the principles or guidelines. The 4 guidelines provide the foundation goals that authors should aim for in order to make content accessible to people with disabilities. The guidelines are necessary but provide the framework and not enough on their own. Without these four principles, content is not accessible to people with disabilities. Principles are necessary but provide the framework and not enough on their own.

Web Site can consist of one Web Site or thousands of Web Sites and terms accessible is individuals with all types, degrees, or combinations of disability, particularly in the cognitive, language, and learning areas. Authors are encouraged to provide alternatives for different types of content, including different multimedia formats, and to ensure that content is accessible, as far as possible, to the widest audience. Web Sites have been in existence since about 1994 and have been proven to conform that Web content is accessible, as far as possible, to the widest audience. Web Sites have been in existence since about 1994 and have been proven to conform that Web content is accessible, as far as possible, to the widest audience.

All of our products should aim for the AA-standard of WCAG in terms of accessibility. This means fulfilling all of the criteria for both A- and AA-levels.

Image source: https://www.w3.org/WAI/WCAG21/quickref/#qr-intro-principles-guidelines

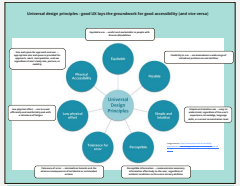
WCAG Success Criteria

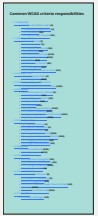
- **Level A** includes the most essential (and least strict) requirements.
- **Level AA** includes additional criteria. Most websites should aim for Level AA conformance.
- **Level AAA** includes the most strict requirements. Some types of content may be unable to meet all Level AAA success criteria.

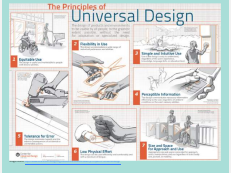
© 2018 TAMUCC-IMT

Responsibilities

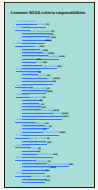
Designers



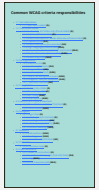




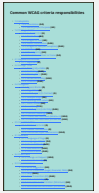
Developers



Marketing & content



PO & PM



QA & testing

Questions board

Click to add

Designers

Universal design principles - good UX lays the groundwork for good accessibility (and vice versa)

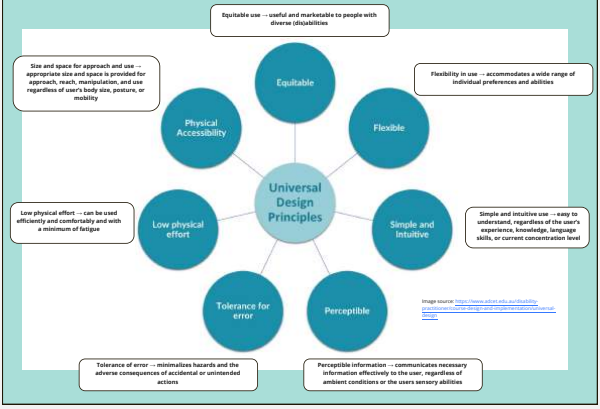


Image source: https://www.adcet.edu.au/disability-practitioners/course-design-and-implementation/universal-design

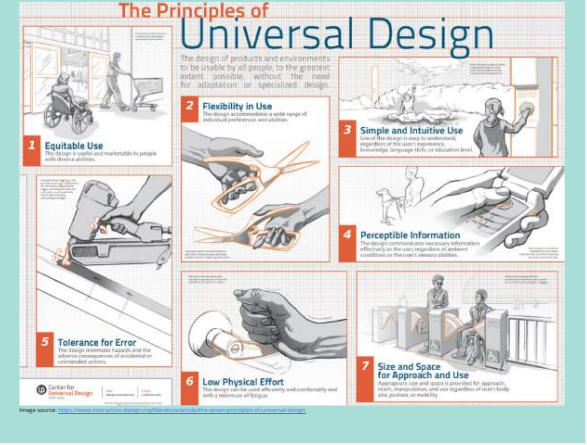
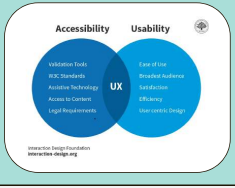


Image source: https://www.interaction-design.org/literature/article/the-seven-principles-of-universal-design

- Accessibility helps with design work
  - Aim to find the simplest possible solution to the needs at hand
- Promotes staff productivity and creativity
  - Design for all vs. fill this criteria
- Designers bear the initial brunt of accessibility efforts by designing elements, components and logic that work for everyone. Especially visual aspects tend to be design tasks



Common WCAG criteria responsibilities

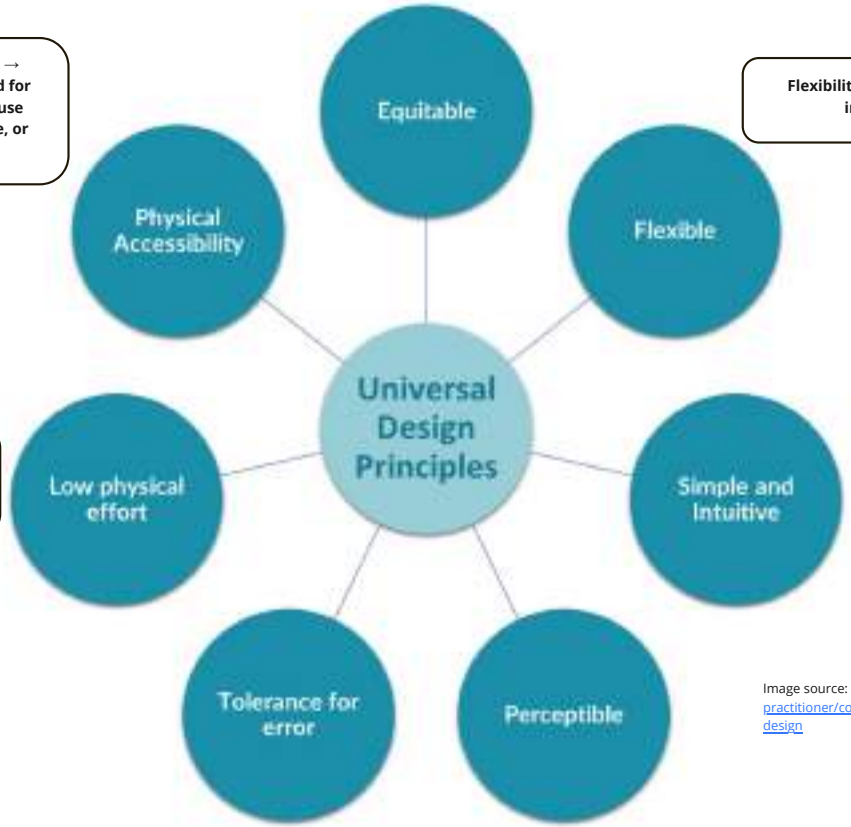
- 1.3 Adaptable
  - 1.3.1 Info and Relationships (A)
  - 1.3.3 Sensory Characteristics (A)
  - 1.3.4 Orientation (AA)
  - 1.3.5 Identify Input Purpose (AA)
- 1.4 Distinguishable
  - 1.4.1 Use of Color (A)
  - 1.4.2 Audio Control (A)
  - 1.4.3 Contrast (Minimum) (AA)
  - 1.4.4 Resize Text (AA)
  - 1.4.6 Contrast (Enhanced) (AAA)
  - 1.4.8 Visual Presentation (AAA)
  - 1.4.10 Reflow (AA)
  - 1.4.11 Non-text Contrast (AA)
  - 1.4.12 Text Spacing (AA)
  - 1.4.13 Content on Hover or Focus (AA)
- 2.2 Enough Time
  - 2.2.1 Timing Adjustable (A)
  - 2.2.4 Interruptions (AAA)
- 2.3 Seizures and Physical Reactions
  - 2.3.3 Animation from Interactions (AAA)
- 2.4 Navigable
  - 2.4.4 Link Purpose (In Context) (A)
  - 2.4.5 Multiple Ways (AA)
  - 2.4.6 Headings and Labels (AA)
  - 2.4.7 Focus Visible (AA)
  - 2.4.8 Location (AAA)
  - 2.4.9 Link Purpose (Link Only) (AAA)
  - 2.4.10 Section Headings (AAA)
  - 2.4.12 Focus Not Obscured (Enhanced) (AAA)
  - 2.4.13 Focus Appearance (AAA)
- 2.5 Input Modalities
  - 2.5.1 Pointer Gestures (A)
  - 2.5.2 Pointer Cancellation (A)
  - 2.5.4 Motion Actuation (A)
  - 2.5.5 Target Size (Enhanced) (AAA)
  - 2.5.6 Concurrent Input Mechanisms (AAA)
  - 2.5.7 Dragging Movements (AA)
  - 2.5.8 Target Size (Minimum) (AA)
- 3.1 Readable
  - 3.1.3 Unusual Words (AAA)
  - 3.1.4 Abbreviations (AAA)
- 3.2 Predictable
  - 3.2.3 Consistent Navigation (AA)
  - 3.2.4 Consistent Identification (AA)
  - 3.2.5 Change on Request (AAA)
  - 3.2.6 Consistent Help (A)
- 3.3 Input Assistance
  - 3.3.1 Error Identification (A)
  - 3.3.2 Labels or Instructions (A)
  - 3.3.3 Error Suggestion (AA)
  - 3.3.4 Error Prevention (Legal, Financial, Data) (AA)
  - 3.3.5 Help (AAA)
  - 3.3.6 Error Prevention (All) (AAA)
- 4.1 Compatible
  - 4.1.3 Status Messages (AA)

Universal design principles - good UX lays the groundwork for good accessibility (and vice versa)

Equitable use → useful and marketable to people with diverse (dis)abilities

Size and space for approach and use → appropriate size and space is provided for approach, reach, manipulation, and use regardless of user's body size, posture, or mobility

Flexibility in use → accommodates a wide range of individual preferences and abilities



Low physical effort → can be used efficiently and comfortably and with a minimum of fatigue

Simple and intuitive use → easy to understand, regardless of the user's experience, knowledge, language skills, or current concentration level

Image source: https://www.adcet.edu.au/disability-practitioner/course-design-and-implementation/universal-design

Tolerance of error → minimalizes hazards and the adverse consequences of accidental or unintended actions

Perceptible information → communicates necessary information effectively to the user, regardless of ambient conditions or the users sensory abilities

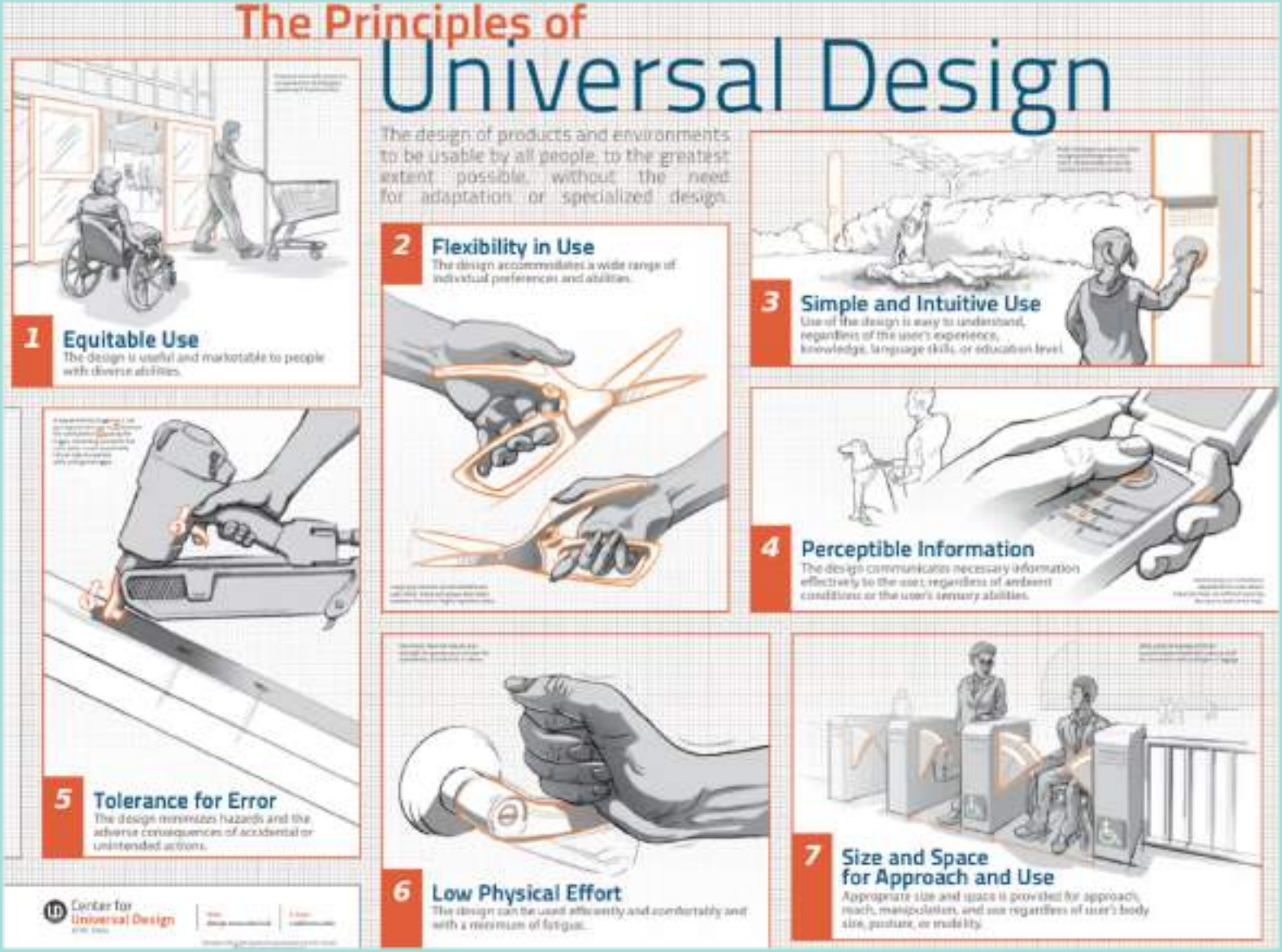
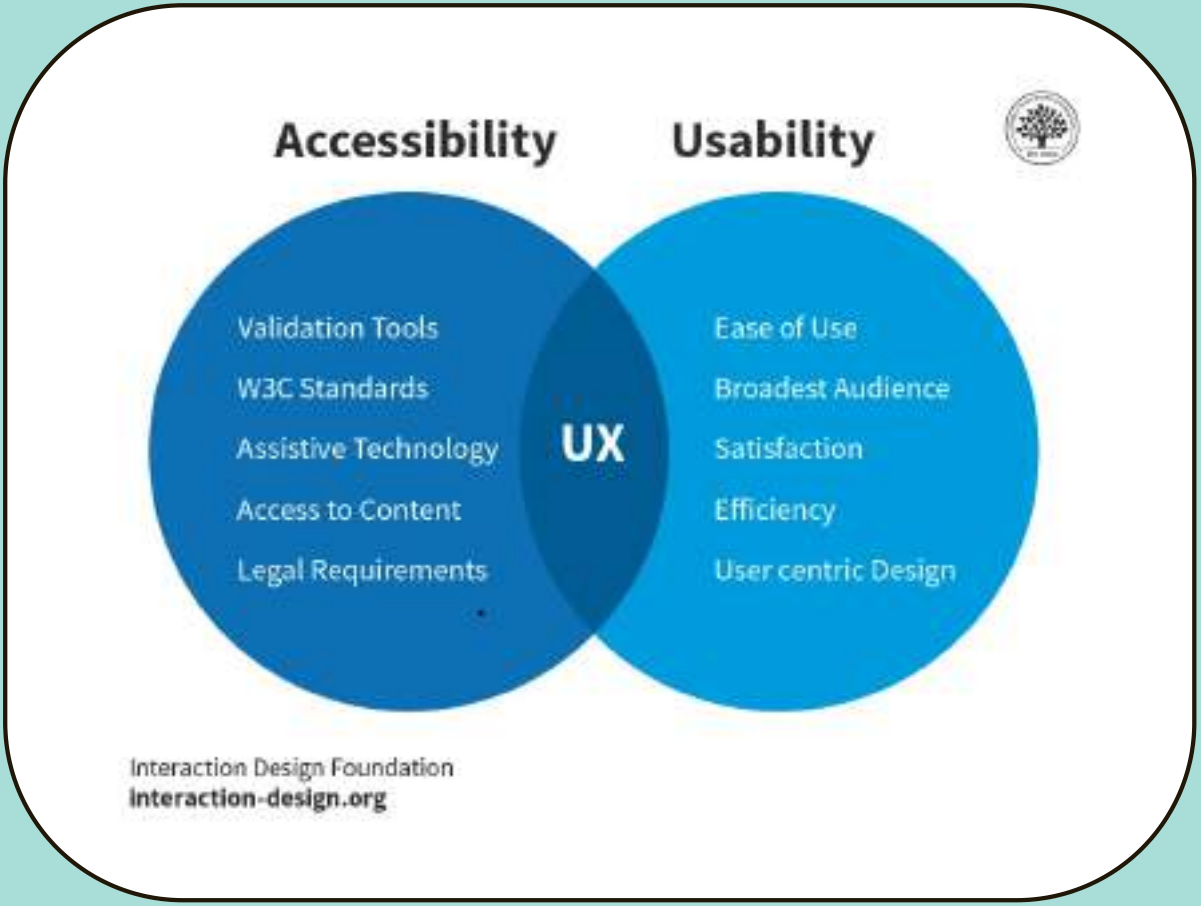


Image source: https://www.interaction-design.org/literature/article/the-seven-principles-of-universal-design

- Accessibility helps with design work
    - Aim to find the simplest possible solution to the needs at hand
- Promotes staff productivity and creativity
    - Design for all vs. fill this criteria
- Designers bear the initial brunt of accessibility efforts by designing elements, components and logic that work for everyone. Especially visual aspects tend to be design tasks



Common WCAG criteria responsibilities

- 1.3 Adaptable
  - 1.3.1 Info and Relationships **(A)**
  - 1.3.3 Sensory Characteristics **(A)**
  - 1.3.4 Orientation **(AA)**
  - 1.3.5 Identify Input Purpose **(AA)**
- 1.4 Distinguishable
  - 1.4.1 Use of Color **(A)**
  - 1.4.2 Audio Control **(A)**
  - 1.4.3 Contrast (Minimum) **(AA)**
  - 1.4.4 Resize Text **(AA)**
  - 1.4.6 Contrast (Enhanced) **(AAA)**
  - 1.4.8 Visual Presentation **(AAA)**
  - 1.4.10 Reflow **(AA)**
  - 1.4.11 Non-text Contrast **(AA)**
  - 1.4.12 Text Spacing **(AA)**
  - 1.4.13 Content on Hover or Focus **(AA)**
- 2.2 Enough Time
  - 2.2.1 Timing Adjustable **(A)**
  - 2.2.4 Interruptions **(AAA)**
- 2.3 Seizures and Physical Reactions
  - 2.3.3 Animation from Interactions **(AAA)**
- 2.4 Navigable
  - 2.4.4 Link Purpose (In Context) **(A)**
  - 2.4.5 Multiple Ways **(AA)**
  - 2.4.6 Headings and Labels **(AA)**
  - 2.4.7 Focus Visible **(AA)**
  - 2.4.8 Location **(AAA)**
  - 2.4.9 Link Purpose (Link Only) **(AAA)**
  - 2.4.10 Section Headings **(AAA)**
  - 2.4.12 Focus Not Obscured (Enhanced) **(AAA)**
  - 2.4.13 Focus Appearance **(AAA)**
- 2.5 Input Modalities
  - 2.5.1 Pointer Gestures **(A)**
  - 2.5.2 Pointer Cancellation **(A)**
  - 2.5.4 Motion Actuation **(A)**
  - 2.5.5 Target Size (Enhanced) **(AAA)**
  - 2.5.6 Concurrent Input Mechanisms **(AAA)**
  - 2.5.7 Dragging Movements **(AA)**
  - 2.5.8 Target Size (Minimum) **(AA)**
- 3.1 Readable
  - 3.1.3 Unusual Words **(AAA)**
  - 3.1.4 Abbreviations **(AAA)**
- 3.2 Predictable
  - 3.2.3 Consistent Navigation **(AA)**
  - 3.2.4 Consistent Identification **(AA)**
  - 3.2.5 Change on Request **(AAA)**
  - 3.2.6 Consistent Help **(A)**
- 3.3 Input Assistance
  - 3.3.1 Error Identification **(A)**
  - 3.3.2 Labels or Instructions **(A)**
  - 3.3.3 Error Suggestion **(AA)**
  - 3.3.4 Error Prevention (Legal, Financial, Data) **(AA)**
  - 3.3.5 Help **(AAA)**
  - 3.3.6 Error Prevention (All) **(AAA)**
- 4.1 Compatible
  - 4.1.3 Status Messages **(AA)**

Developers

- Flawless technical implementation
  - Logical and semantic code that follows standards and WCAG guidelines, and works well with variety of devices and technologies
- Accessibility helps in daily development work
  - Robust foundation, clear goals and good technically sustainable solutions -> better Developer experience
- Ensure basics
  - If your website/application can be used with just keyboard and screen reader, you will cover the majority of accessibility use cases

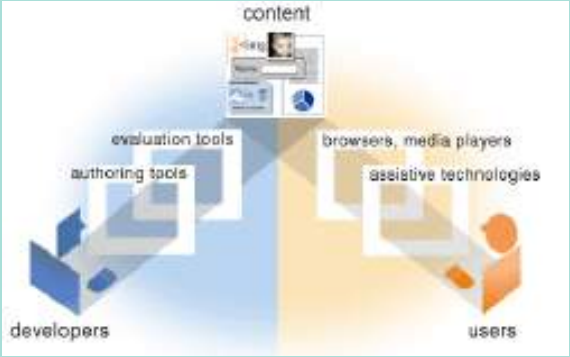


Image source: https://skillcrush.com/blog/what-is-web-accessibility/

Common WCAG criteria responsibilities

- 1.1 Text alternatives
  - 1.1.1 Non-text Content **(A)**
- 1.3 Adaptable
  - 1.3.1 Info and Relationships **(A)**
  - 1.3.2 Meaningful Sequence **(A)**
  - 1.3.5 Identify Input Purpose **(AA)**
  - 1.3.6 Identify Purpose **(AAA)**
- 1.4 Distinguishable
  - 1.4.4 Resize Text **(AA)**
  - 1.4.5 Images of Text **(AA)**
  - 1.4.8 Visual Presentation **(AAA)**
  - 1.4.9 Images of Text (No Exception) **(AAA)**
  - 1.4.10 Reflow **(AA)**
  - 1.4.12 Text Spacing **(AA)**
  - 1.4.13 Content on Hover or Focus **(AA)**
- 2.1 Keyboard Accessible
  - 2.1.1 Keyboard **(A)**
  - 2.1.2 No Keyboard Trap **(A)**
  - 2.1.3 Keyboard (No Exception) **(AAA)**
  - 2.1.4 Character Key Shortcuts **(A)**
- 2.2 Enough Time
  - 2.2.1 Timing Adjustable **(A)**
  - 2.2.5 Re-authenticating **(AAA)**
- 2.4 Navigable
  - 2.4.1 Bypass Blocks **(A)**
  - 2.4.2 Page Titled **(A)**
  - 2.4.3 Focus Order **(A)**
  - 2.4.7 Focus Visible **(AA)**
  - 2.4.9 Link Purpose (Link Only) **(AAA)**
  - 2.4.11 Focus Not Obscured (Minimum) **(AAA)**
  - 2.4.12 Focus Not Obscured (Enhanced) **(AAA)**
- 2.5 Input Modalities
  - 2.5.1 Pointer Gestures **(A)**
  - 2.5.2 Pointer Cancellation **(A)**
  - 2.5.3 Label in Name **(A)**
  - 2.5.6 Concurrent Input Mechanisms **(AAA)**
- 3.1 Readable
  - 3.1.1 Language of Page **(A)**
  - 3.1.2 Language of Parts **(AA)**
- 3.2 Predictable
  - 3.2.1 On Focus **(A)**
  - 3.2.2 On Input **(A)**
  - 3.2.4 Consistent Identification **(AA)**
- 3.3 Input Assistance
  - 3.3.1 Error Identification **(A)**
  - 3.3.4 Error Prevention (Legal, Financial, Data) **(AA)**
  - 3.3.6 Error Prevention (All) **(AAA)**
  - 3.3.7 Redundant Entry **(A)**
- 4.1 Compatible
  - 4.1.2 Name, Role, Value **(A)**
  - 4.1.3 Status Messages **(AA)**

- Flawless technical implementation
    - Logical and semantic code that follows standards and WCAG guidelines, and works well with variety of devices and technologies
- Accessibility helps in daily development work
    - Robust foundation, clear goals and good technically sustainable solutions -> better Developer experience
- Ensure basics
    - If your website/application can be used with just keyboard and screen reader, you will cover the majority of accessibility use cases

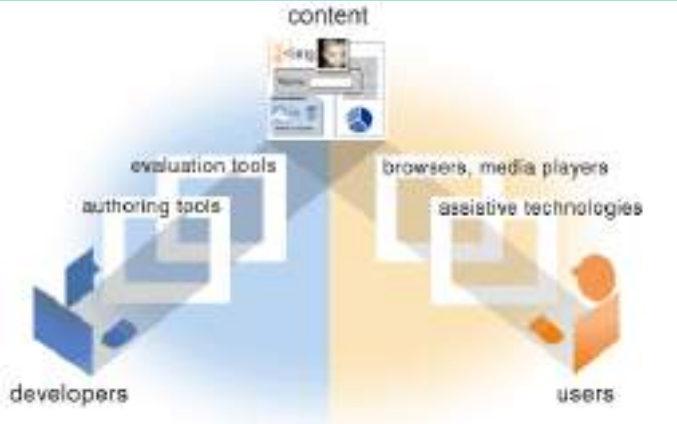


Image source: https://skillcrush.com/blog/what-is-web-accessibility/

Common WCAG criteria responsibilities

- 1.1 Text alternatives
  - 1.1.1 Non-text Content **(A)**
- 1.3 Adaptable
  - 1.3.1 Info and Relationships **(A)**
  - 1.3.2 Meaningful Sequence **(A)**
  - 1.3.5 Identify Input Purpose **(AA)**
  - 1.3.6 Identify Purpose **(AAA)**
- 1.4 Distinguishable
  - 1.4.4 Resize Text **(AA)**
  - 1.4.5 Images of Text **(AA)**
  - 1.4.8 Visual Presentation **(AAA)**
  - 1.4.9 Images of Text (No Exception) **(AAA)**
  - 1.4.10 Reflow **(AA)**
  - 1.4.12 Text Spacing **(AA)**
  - 1.4.13 Content on Hover or Focus **(AA)**
- 2.1 Keyboard Accessible
  - 2.1.1 Keyboard **(A)**
  - 2.1.2 No Keyboard Trap **(A)**
  - 2.1.3 Keyboard (No Exception) **(AAA)**
  - 2.1.4 Character Key Shortcuts **(A)**
- 2.2 Enough Time
  - 2.2.1 Timing Adjustable **(A)**
  - 2.2.5 Re-authenticating **(AAA)**
- 2.4 Navigable
  - 2.4.1 Bypass Blocks **(A)**
  - 2.4.2 Page Titled **(A)**
  - 2.4.3 Focus Order **(A)**
  - 2.4.7 Focus Visible **(AA)**
  - 2.4.9 Link Purpose (Link Only) **(AAA)**
  - 2.4.11 Focus Not Obscured (Minimum) **(AAA)**
  - 2.4.12 Focus Not Obscured (Enhanced) **(AAA)**
- 2.5 Input Modalities
  - 2.5.1 Pointer Gestures **(A)**
  - 2.5.2 Pointer Cancellation **(A)**
  - 2.5.3 Label in Name **(A)**
  - 2.5.6 Concurrent Input Mechanisms **(AAA)**
- 3.1 Readable
  - 3.1.1 Language of Page **(A)**
  - 3.1.2 Language of Parts **(AA)**
- 3.2 Predictable
  - 3.2.1 On Focus **(A)**
  - 3.2.2 On Input **(A)**
  - 3.2.4 Consistent Identification **(AA)**
- 3.3 Input Assistance
  - 3.3.1 Error Identification **(A)**
  - 3.3.4 Error Prevention (Legal, Financial, Data) **(AA)**
  - 3.3.6 Error Prevention (All) **(AAA)**
  - 3.3.7 Redundant Entry **(A)**
- 4.1 Compatible
  - 4.1.2 Name, Role, Value **(A)**
  - 4.1.3 Status Messages **(AA)**

Marketing & content

- Accessible content
  - If your content is invisible or incomprehensible to users, for example info is poorly tagged or described, why have it at all?
- Selling points
  - Attracts more users and user groups
  - Easy to approach UI
  - Ability to sell to a wider and more varied environment
  - Customer retention is easier
  - Good reputation
  - Trust
- Marketing points
  - Supports digital marketing
  - Decreases bounce rate
  - Search engine optimization is easier

Image source: https://reciteme.com/au/news/how-does-web-accessibility-impact-marketing-outcomes/

Common WCAG criteria responsibilities

- 1.1 Text alternatives
  - 1.1.1 Non-text Content **(A)**
- 1.2 Time-based Media
  - 1.2.1 Audio-only and Video-only (Prerecorded) **(A)**
  - 1.2.2 Captions (Prerecorded) **(A)**
  - 1.2.3 Audio Description or Media Alternative (Prerecorded) **(A)**
  - 1.2.4 Captions (Live) **(AA)**
  - 1.2.5 Audio Description (Prerecorded) **(AA)**
  - 1.2.6 Sign Language (Prerecorded) **(AAA)**
  - 1.2.7 Extended Audio Description (Prerecorded) **(AAA)**
  - 1.2.8 Media Alternative (Prerecorded) **(AAA)**
  - 1.2.9 Audio-only (Live) **(AAA)**
- 1.4 Distinguishable
  - 1.4.2 Audio Control **(A)**
  - 1.4.3 Contrast (Minimum) **(AA)**
  - 1.4.5 Images of Text **(AA)**
  - 1.4.6 Contrast (Enhanced) **(AAA)**
  - 1.4.7 Low or No Background Audio **(AAA)**
  - 1.4.9 Images of Text (No Exception) **(AAA)**
  - 1.4.11 Non-text Contrast **(AA)**
- 2.2 Enough Time
  - 2.2.1 Timing Adjustable **(A)**
  - 2.2.2 Pause, Stop, Hide **(A)**
  - 2.2.3 No Timing **(AAA)**
  - 2.2.4 Interruptions **(AAA)**
- 2.3 Seizures and Physical Reactions
  - 2.3.1 Three Flashes or Below Threshold **(A)**
  - 2.3.2 Three Flashes **(AAA)**
- 2.4 Navigable
  - 2.4.2 Page Titled **(A)**
  - 2.4.4 Link Purpose (In Context) **(A)**
  - 2.4.6 Headings and Labels **(AA)**
  - 2.4.9 Link Purpose (Link Only) **(AAA)**
  - 2.4.10 Section Headings **(AAA)**
- 3.1 Readable
  - 3.1.4 Abbreviations **(AAA)**
  - 3.1.5 Reading Level **(AAA)**
  - 3.2.4 Consistent Identification **(AA)**
- 3.2 Predictable
  - 3.2.6 Consistent Help **(A)**
- 3.3 Input Assistance
  - 3.3.2 Labels or Instructions **(A)**
  - 3.3.4 Error Prevention (Legal, Financial, Data) **(AA)**
  - 3.3.5 Help **(AAA)**
  - 3.3.6 Error Prevention (All) **(AAA)**

- Accessible content
    - If your content is invisible or incomprehensible to users, for example info is poorly tagged or described, why have it at all?
- Selling points
    - Attracts more users and user groups
    - Easy to approach UI
    - Ability to sell to a wider and more varied environment
    - Customer retention is easier
    - Good reputation
    - Trust
- Marketing points
    - Supports digital marketing
    - Decreases bounce rate
    - Search engine optimization is easier

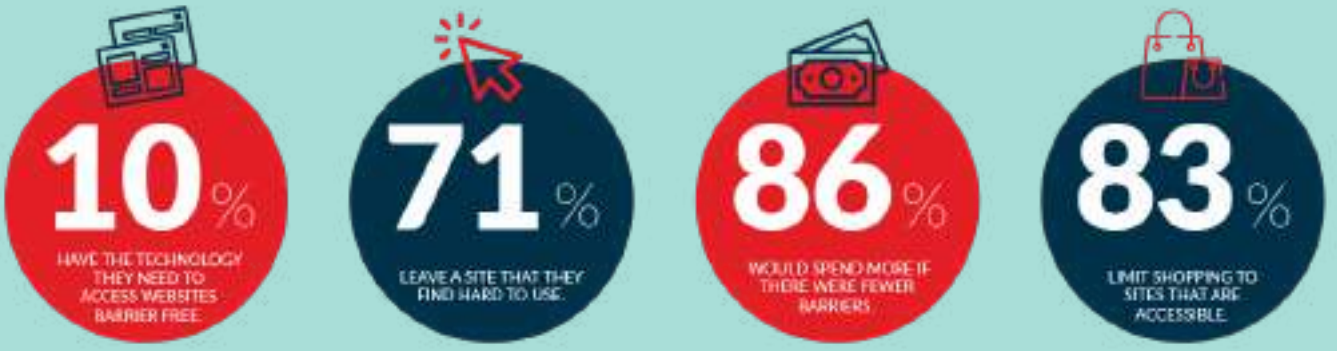


Image source: https://reciteme.com/au/news/how-does-web-accessibility-impact-marketing-outcomes/

Common WCAG criteria responsibilities

- 1.1 Text alternatives
  - 1.1.1 Non-text Content **(A)**
- 1.2 Time-based Media
  - 1.2.1 Audio-only and Video-only (Prerecorded) **(A)**
  - 1.2.2 Captions (Prerecorded) **(A)**
  - 1.2.3 Audio Description or Media Alternative (Prerecorded) **(A)**
  - 1.2.4 Captions (Live) **(AA)**
  - 1.2.5 Audio Description (Prerecorded) **(AA)**
  - 1.2.6 Sign Language (Prerecorded) **(AAA)**
  - 1.2.7 Extended Audio Description (Prerecorded) **(AAA)**
  - 1.2.8 Media Alternative (Prerecorded) **(AAA)**
  - 1.2.9 Audio-only (Live) **(AAA)**
- 1.4 Distinguishable
  - 1.4.2 Audio Control **(A)**
  - 1.4.3 Contrast (Minimum) **(AA)**
  - 1.4.5 Images of Text **(AA)**
  - 1.4.6 Contrast (Enhanced) **(AAA)**
  - 1.4.7 Low or No Background Audio **(AAA)**
  - 1.4.9 Images of Text (No Exception) **(AAA)**
  - 1.4.11 Non-text Contrast **(AA)**
- 2.2 Enough Time
  - 2.2.1 Timing Adjustable **(A)**
  - 2.2.2 Pause, Stop, Hide **(A)**
  - 2.2.3 No Timing **(AAA)**
  - 2.2.4 Interruptions **(AAA)**
- 2.3 Seizures and Physical Reactions
  - 2.3.1 Three Flashes or Below Threshold **(A)**
  - 2.3.2 Three Flashes **(AAA)**
- 2.4 Navigable
  - 2.4.2 Page Titled **(A)**
  - 2.4.4 Link Purpose (In Context) **(A)**
  - 2.4.6 Headings and Labels **(AA)**
  - 2.4.9 Link Purpose (Link Only) **(AAA)**
  - 2.4.10 Section Headings **(AAA)**
- 3.1 Readable
  - 3.1.4 Abbreviations **(AAA)**
  - 3.1.5 Reading Level **(AAA)**
  - 3.2.4 Consistent Identification **(AA)**
- 3.2 Predictable
  - 3.2.6 Consistent Help **(A)**
- 3.3 Input Assistance
  - 3.3.2 Labels or Instructions **(A)**
  - 3.3.4 Error Prevention (Legal, Financial, Data) **(AA)**
  - 3.3.5 Help **(AAA)**
  - 3.3.6 Error Prevention (All) **(AAA)**

PO & PM

- POs/PMs serve as an intermediary between all of the other roles and should have a general awareness of most accessibility acceptance criteria and who should focus on what
- Their responsibilities also include the logic of the product/service and functionality relating to that
- They share responsibilities with all other roles, but also have some of their own

Image source: https://digital.gov/guides/accessibility-for-teams/product-management/

Common WCAG criteria responsibilities

- 1.3 Adaptable
  - 1.3.4 Orientation **(AA)**
  - 1.3.5 Identify Input Purpose **(AA)**
- 1.4 Distinguishable
  - 1.4.2 Audio Control **(A)**
  - 1.4.4 Resize Text **(AA)**
  - 1.4.5 Images of Text **(AA)**
  - 1.4.8 Visual Presentation **(AAA)**
  - 1.4.9 Images of Text (No Exception) **(AAA)**
  - 1.4.10 Reflow **(AA)**
  - 1.4.12 Text Spacing **(AA)**
  - 1.4.13 Content on Hover or Focus **(AA)**
- 2.1 Keyboard Accessible
  - 2.1.1 Keyboard **(A)**
- 2.2 Enough Time
  - 2.2.1 Timing Adjustable **(A)**
  - 2.2.3 No Timing **(AAA)**
  - 2.2.4 Interruptions **(AAA)**
  - 2.2.5 Re-authenticating **(AAA)**
  - 2.2.6 Timeouts **(AAA)**
- 2.4 Navigable
  - 2.4.1 Bypass Blocks **(A)**
  - 2.4.2 Page Titled **(A)**
  - 2.4.4 Link Purpose (In Context) **(A)**
  - 2.4.5 Multiple Ways **(AA)**
  - 2.4.6 Headings and Labels **(AA)**
  - 2.4.7 Focus Visible **(AA)**
  - 2.4.8 Location **(AAA)**
  - 2.4.9 Link Purpose (Link Only) **(AAA)**
  - 2.4.10 Section Headings **(AAA)**
  - 2.4.11 Focus Not Obscured (Minimum) **(AAA)**
  - 2.4.12 Focus Not Obscured (Enhanced) **(AAA)**
- 2.5 Input Modalities
  - 2.5.1 Pointer Gestures **(A)**
  - 2.5.2 Pointer Cancellation **(A)**
  - 2.5.6 Concurrent Input Mechanisms **(AAA)**
- 3.1 Readable
  - 3.1.1 Language of Page **(A)**
  - 3.1.2 Language of Parts **(AA)**
  - 3.1.3 Unusual Words **(AAA)**
  - 3.1.4 Abbreviations **(AAA)**
  - 3.1.6 Pronunciation **(AAA)**
- 3.2 Predictable
  - 3.2.5 Change on Request **(AAA)**
  - 3.2.6 Consistent Help **(A)**
- 3.3 Input Assistance
  - 3.3.3 Error Suggestion **(AA)**
  - 3.3.4 Error Prevention (Legal, Financial, Data) **(AA)**
  - 3.3.5 Help **(AAA)**
  - 3.3.6 Error Prevention (All) **(AAA)**
  - 3.3.7 Redundant Entry **(A)**
  - 3.3.8 Accessible Authentication (Minimum) **(AA)**
  - 3.3.9 Accessible Authentication (Enhanced) **(AAA)**

- POs/PMs serve as an intermediary between all of the other roles and should have a general awareness of most accessibility acceptance criteria and who should focus on what
- Their responsibilities also include the logic of the product/service and functionality relating to that
- They share responsibilities with all other roles, but also have some of their own

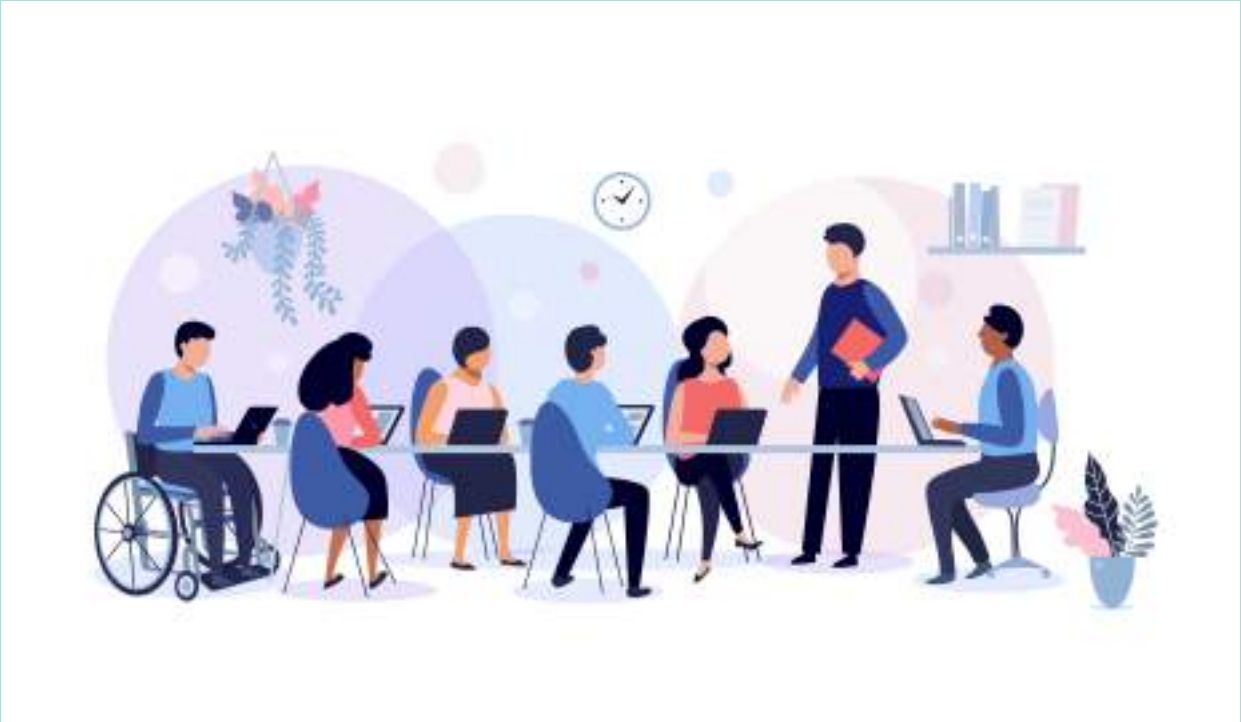


Image source: https://digital.gov/guides/accessibility-for-teams/product-management/

Common WCAG criteria responsibilities

- 1.3 Adaptable
  - 1.3.4 Orientation **(AA)**
  - 1.3.5 Identify Input Purpose **(AA)**
- 1.4 Distinguishable
  - 1.4.2 Audio Control **(A)**
  - 1.4.4 Resize Text **(AA)**
  - 1.4.5 Images of Text **(AA)**
  - 1.4.8 Visual Presentation **(AAA)**
  - 1.4.9 Images of Text (No Exception) **(AAA)**
  - 1.4.10 Reflow **(AA)**
  - 1.4.12 Text Spacing **(AA)**
  - 1.4.13 Content on Hover or Focus **(AA)**
- 2.1 Keyboard Accessible
  - 2.1.1 Keyboard **(A)**
- 2.2 Enough Time
  - 2.2.1 Timing Adjustable **(A)**
  - 2.2.3 No Timing **(AAA)**
  - 2.2.4 Interruptions **(AAA)**
  - 2.2.5 Re-authenticating **(AAA)**
  - 2.2.6 Timeouts **(AAA)**
- 2.4 Navigable
  - 2.4.1 Bypass Blocks **(A)**
  - 2.4.2 Page Titled **(A)**
  - 2.4.4 Link Purpose (In Context) **(A)**
  - 2.4.5 Multiple Ways **(AA)**
  - 2.4.6 Headings and Labels **(AA)**
  - 2.4.7 Focus Visible **(AA)**
  - 2.4.8 Location **(AAA)**
  - 2.4.9 Link Purpose (Link Only) **(AAA)**
  - 2.4.10 Section Headings **(AAA)**
  - 2.4.11 Focus Not Obscured (Minimum) **(AAA)**
  - 2.4.12 Focus Not Obscured (Enhanced) **(AAA)**
- 2.5 Input Modalities
  - 2.5.1 Pointer Gestures **(A)**
  - 2.5.2 Pointer Cancellation **(A)**
  - 2.5.6 Concurrent Input Mechanisms **(AAA)**
- 3.1 Readable
  - 3.1.1 Language of Page **(A)**
  - 3.1.2 Language of Parts **(AA)**
  - 3.1.3 Unusual Words **(AAA)**
  - 3.1.4 Abbreviations **(AAA)**
  - 3.1.6 Pronunciation **(AAA)**
- 3.2 Predictable
  - 3.2.5 Change on Request **(AAA)**
  - 3.2.6 Consistent Help **(A)**
- 3.3 Input Assistance
  - 3.3.3 Error Suggestion **(AA)**
  - 3.3.4 Error Prevention (Legal, Financial, Data) **(AA)**
  - 3.3.5 Help **(AAA)**
  - 3.3.6 Error Prevention (All) **(AAA)**
  - 3.3.7 Redundant Entry **(A)**
  - 3.3.8 Accessible Authentication (Minimum) **(AA)**
  - 3.3.9 Accessible Authentication (Enhanced) **(AAA)**

QA & testing

- Testers/QA is uniquely positioned in that no particular criteria is their sole responsibility, yet most of them are.
- Especially designers' and developers' responsibilities are most commonly QA's responsibilities as well.
- Accessibility testing coverage should be agreed on and then conducted accordingly.
- Level A should ALWAYS be tested for. AA should be the goal in most products, AAA is a luxury that you can aim for if you have the capabilities and possibilities to do so.
- QA should be the last line of testing that catches what fell through the gaps. Everyone in the development cycle is responsible for testing for accessibility

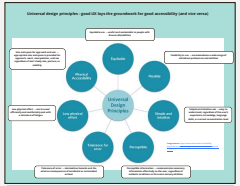
Image source: https://www.nagarro.com/en/blog/power-of-accessibility-testing

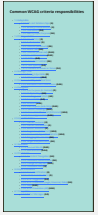
- Testers/QA is uniquely positioned in that no particular criteria is their sole responsibility, yet most of them are.
- Especially designers' and developers' responsibilities are most commonly QA's responsibilities as well.
- Accessibility testing coverage should be agreed on and then conducted accordingly.
- Level A should ALWAYS be tested for. AA should be the goal in most products, AAA is a luxury that you can aim for if you have the capabilities and possibilities to do so.
- QA should be the last line of testing that catches what fell through the gaps. Everyone in the development cycle is responsible for testing for accessibility

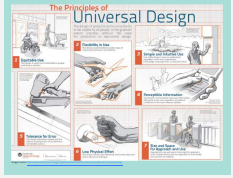
Image source: https://www.nagarro.com/en/blog/power-of-accessibility-testing

Responsibilities

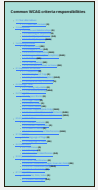
Designers



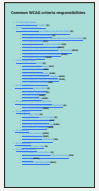




Developers



Marketing & content



PO & PM

QA & testing

Questions board

Accessibility in our products

Under this branch you can find guidance on what to do with your product in terms of accessibility and the current accessibility status of all our products.

To-Do -list for all products

Accessibility Statement

- Key requirement by the Accessibility compliance (EU and national)
- Must contain (in Finland):
  - General evaluation of the level of compliance
  - List of deficiencies

Accessibility Champions

- A person already working in a product team who is committed to sharing the message of accessibility inside their product organization
- High and moderate risk products need 2: 1 UX & 1 Dev

Accessibility Plan

- A report of the current accessibility issues.
- Estimates and costs, general plan of action to become accessible

Visma Accessibility Self-Assessment

- Evaluate product towards Visma Accessibility Maturity Index
- Part of the Accessibility plan, tool to evaluate and understand problems

Step 1: Conduct an accessibility review for your product.

- To start any accessibility journey, you have to realize where you are. For this, you need an accessibility assessment. You have 2 options on how to accomplish it: have someone internally use a template to conduct the analysis or hire an outside party to make the assessment.
- Document all findings during the assessment process so the form, location and magnitude of different issues can be revisited once repairs begin.

Step 2: Create the Accessibility Statement for your product.

- The Accessibility Statement is the core part of the compliance
- Contains key information about the product accessibility as well as containing necessary feedback channels and actions for a user
- According to the legislator the Accessibility Statement must contain the following information:
  - Level of compliance of a product
    - Compliant
    - Mostly compliant
    - Mostly not compliant
  - Parts or features of a product that are not compliant
    - WCAG categorization is sufficient
  - Date and creator of the assessment
    - Internal or external
  - Feedback channel for accessibility issues
    - Must to contain information on who or what entity inside the organization receives the feedback
  - Information on how to contact the national supervisor (AVI before 1.1.2025, Traficom after)
    - To request an audit or to make a complaint

Step 3: Select 2 Accessibility Champions and make a Accessibility Plan

- Visma Group requirements for all moderate risk products
  - Basically every product that needs an Accessibility Statement as per the national supervisor (outside Norway)
- Accessibility Champions are volunteers with a task to to advocate accessibility for a product and ensure that the knowledge and readiness of accessibility is at a adequate level inside product
  - 2 required per product: 1 developer and 1 designer
- Accessibility plan is contains a set of steps necessary to improve product accessibility
  - A report of the current accessibility issues.
  - Estimated time and cost to become compliant
  - Milestones and a reasonable timeline for when your product can become compliant

Image source: https://www.vecteezy.com/vector-art/9-free-vector-accessibility-illustration-pack-for-ui-ux-and-other-uses

Step 4: Improve accessibility

- Follow the Accessibility plan
- Repair issues
- Educate employees
- Accessibility as a Definition of Done
- Accessibility testing and verifying
- Yearly audits, internal of external
- Make steady improvements

Image source: https://www.vecteezy.com/vector-art/21-free-vector-accessibility-illustration-pack-for-ui-ux-and-other-uses

Accessibility in our products

Under this branch you can find guidance on what to do with your product in terms of accessibility and the current accessibility status of all our products.

To-Do -list for all products

Accessibility Statement

- Key requirement by the Accessibility compliance (EU and national)
- Must contain (in Finland):
    - General evaluation of the level of compliance
    - List of deficiencies

Accessibility Champions

- A person already working in a product team who is committed to sharing the message of accessibility inside their product organization
- High and moderate risk products need 2: 1 UX & 1 Dev

Accessibility Plan

- A report of the current accessibility issues.
- Estimates and costs, general plan of action to become accessible

Visma Accessibility Self-Assessment

- Evaluate product towards Visma Accessibility Maturity Index
- Part of the Accessibility plan, tool to evaluate and understand problems

Step 1: Conduct an accessibility review for your product.

- To start any accessibility journey, you have to realize where you are. For this, you need an accessibility assessment. You have 2 options on how to accomplish it: have someone internally use a template to conduct the analysis or hire an outside party to make the assessment.
- Document all findings during the assessment process so the form, location and magnitude of different issues can be revisited once repairs begin.

Step 2: Create the Accessiblity Statement for your product.

- The Accessibility Statement is the core part of the compliance
- Contains key information about the product accessibility as well as containing necessary feedback channels and actions for a user
- According to the legislator the Accessibility Statement must contain the following information:
    - Level of compliance of a product
        - Compliant
        - Mostly compliant
        - Mostly not compliant
    - Parts or features of a product that are not compliant
        - WCAG categorization is sufficient
    - Date and creator of the assessment
        - Internal or external
    - Feedback channel for accessibility issues
        - Must to contain information on who or what entity inside the organization receives the feedback
    - Information on how to contact the national supervisor (AVI before 1.1.2025, Traficom after)
        - To request an audit or to make a complaint

Step 3: Select 2 Accessibility Champions and make a Accessibility Plan

- Visma Group requirements for all moderate risk products
    - Basically every product that needs an Accessibility Statement as per the national supervisor (outside Norway)
- Accessibility Champions are volunteers with a task is to advocate accessibility for a product and ensure that the knowledge and readiness of accessibility is at a adequate level inside product
    - 2 required per product: 1 developer and 1 designer
- Accessibility plan is contains a set of steps necessary to improve product accessibility
    - A report of the current accessibility issues.
    - Estimated time and cost to become compliant
    - Milestones and a reasonable timeline for when your product can become compliant

Image source: https://accessibility.deque.com/how-to-champion-accessibility-and-make-systemic-change

Step 4: Improve accessibility

- Follow the Accessibility plan
- Repair issues
- Educate employees
- Accessibility as a Definition of Done
- Accessibility testing and verifying
- Yearly audits, internal of external
- Make steady improvements

Image source: https://www.soundhound.com/voice-ai-blog/how-voice-assistants-improve-accessibility/

Accessibility in our products

Under this branch you can find guidance on what to do with your product in terms of accessibility and the current accessibility status of all our products.

To-Do -list for all products

Accessibility Statement
- Key requirement by the Accessibility compliance (EU and national)
- Must contain (in Finland)
  - General evaluation of the level of compliance
  - List of deficencies

Accessibility Champions
- A person already working in a product team who is committed to sharing the message of accessibility inside their product organization
- High and moderase risk products need 2: 1 UX & 1 Dev

Accessibility Plan
- A report of the current accessibility issues.
- Estimates and costs, general plan of action to become accessible

Visma Accessibility Self-Assessment
- Evaluate product towards Visma Accessibility Maturity Index
- Part of the Accessibility plan, tool to evaluate and understand problems

Step 1: Conduct an accessibility review for your product.
- To start any accessibility journey, you have to realize where you are. For this, you need an accessibility assessment. You have 2 options on how to accomplish it: have someone internally use a template to conduct the analysis or hire an outside party to make the assessment.
- Document all findings during the assessment process so the form, location and magnitude of different issues can be revisited once repairs begin.

Step 2: Create the Accessibility Statement for your product.

- The Accessibility Statement is the core part of the compliance
- Contains key information about the product accessibility as well as containing necessary feedback channels and actions for a user
- According to the legislator the Accessibility Statement must contain the following information
  - Level of compliance of a product
    - Compliant
    - Mostly compliant
    - Mostly not compliant
  - Parts or features of a product that are not compliant
    - WCAG categorization is sufficient
  - Date and creator of the assessment
    - Internal or external
  - Feedback channel for accessibility issues
    - Must to contain information on who or what entity inside the organization receives the feedback
  - Information on how to contact the national supervisor (AVI before 1.1.2025, Traficom after)
    - To request an audit or to make a complaint

Step 3: Select 3 Accessibility Champions and make a Accessibility Plan
- Visma Group requirements for all moderate risk products
  - Basically every product that needs an Accessibility Statement as per the national supervisor (outside Norway)
- Accessibility Champions are volunteers with a task is to advocate accessibility for a product and ensure that the knowledge and readiness of accessibility is at a adequate level inside product
  - 2 required per product: 1 developer and 1 designer
- Accessibility plan is contains a set of steps necessary to improve product accessibility
  - A report of the current accessibility issues.
  - Estimated time and cost to become compliant
  - Milestones and a reasonable timeline for when your product can become compliant

Image source: https://www.vecteezy.com/vector-art/9-accessibility-icon-design-on-white-background-vector-illustration-with-editable-stroke-vector-illustration-vector-illustration-with-editable-stroke-vector-illustration-vector-illustration-with-editable-stroke-vector-illustration

Where to learn more?

W3 digital accessibility foundations free course: https://www.w3.org/WAI/courses/foundations-course/#video-excerpts-from-the-course

Axe-con accessibility conference. Free and with a massive amount of good videos: https://www.deque.com/axe-con/

Udemy - WCAG 2.1 / 2.2 Simplified With Examples

Udemy - How To Design for Accessibility: for UX Designers (WCAG 2.2)

Udemy (or O'Reilly or whatever other learning platforms we have) in general, you can search for courses on this or other topics

Barclays | Busting Accessibility Myths - https://www.youtube.com/watch?v=_1yGFn7OIDY&list=PLecqH2uhOR0Zb31X7hh5BzWJv4KGLnuUy&index=16

WCAG 2 Overview - https://www.w3.org/WAI/standards-guidelines/wcag/

Accessibility Responsibility Breakdown - https://www.w3.org/community/wai-engage/wiki/Accessibility_Responsibility_Breakdown

Universal Design and Accessibility - https://www.section508.gov/develop/universal-design/

The European Accessibility Act: Empowering Inclusion in Digital Environments: https://medium.com/@norbert-pap/the-european-accessibility-act-empowering-inclusion-in-digital-environments-c5d58f2d854f

Accessibility presentation on color and color contrast: https://docs.google.com/presentation/d/1HvmM142_P3a8j-PWDdb1HzVE_D3W3vN1IOuYExVrm00/edit?usp=sharing

Video of the presentation: https://drive.google.com/file/d/1pO0GXplkbLHiQBr-SAxTrqxeN6CHXEAt/view

EAA Accessibility Checklist https://hello.futurice.com/hubfs/Futurice_Accessibility_Checklist.pdf?utm_campaign=fi-fs1-accessibility-24&utm_medium=email&_hsenc=p2ANqtz-98BH3-IoKiPu_nlD5xUtjoLD2dmOf1NkOuosBukZa3XiWLStEj1QQm-nmzXPKkdzPe2mgU5a4Ku5caDcfN4CA28GMLiIaVs6AcER0wHuDICEAWVaQ&_hsmi=98724457&utm_content=98685265&utm_source=hs_email

How to make people care about accessibility (Axecon 2026): https://youtu.be/RcCblIo1eQ?si=ZH9or9BUPXRIFjMm

Tool repository

Stark Contrast & Accessibility checker for Figma
- https://www.figma.com/community/plugin/732603254453395948/stark-contrast-accessibility-checker

Color Blind plugin for Figma
- https://www.figma.com/community/plugin/733343906244951586/color-blind

WAVE accessibility evaluation browser tool
- https://chromewebstore.google.com/detail/wave-evaluation-tool/jbbplnpkjmmeebjpijfedlgcdilocofh?pli=1

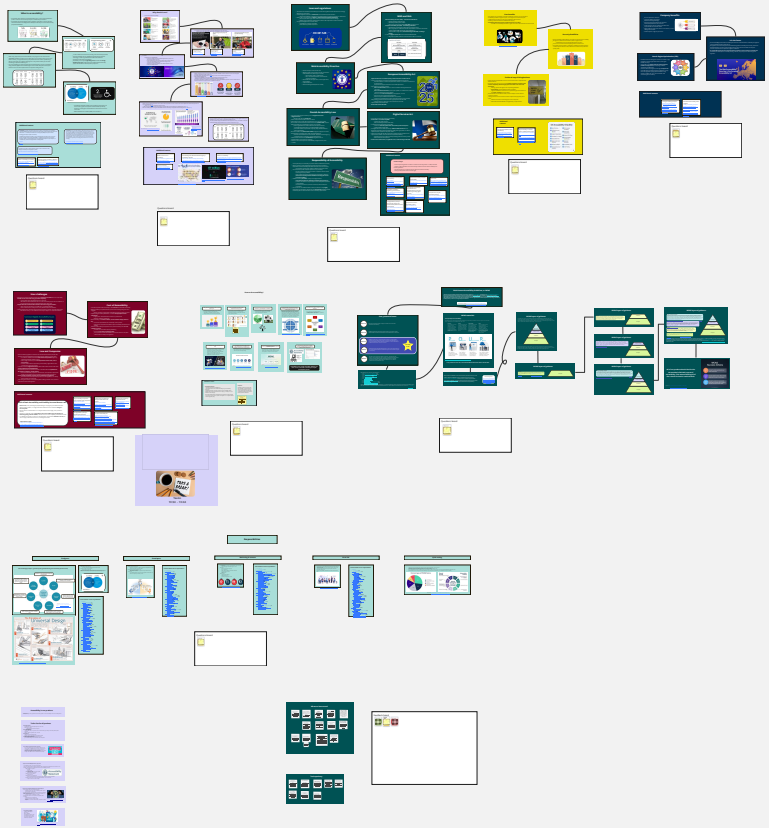
Text Space Editor - extension for Chrome
https://chromewebstore.google.com/detail/text-spacing-editor/amnelgbfbdlfjeaobejkfmjjnmeddaoj

Colour Contrast Checker - extension for Chrome
https://chromewebstore.google.com/detail/colour-contrast-checker/nmmjeclfkgjdomacpcflgdkgpphpmnfe

A11Y - Color blindness empathy test - extension for Chrome
https://chromewebstore.google.com/detail/a11y-color-blindness-empa/idphhflanmeibmjgaciaadkmjebljhcc

Visma Accessibility Self-assessment - Confluence project
https://miro.com/app/board/uXjVKV2HBeo=/?moveToWidget=3458764594686702571&cot=14

Pixel measurement - extension for Chrome
https://chromewebstore.google.com/detail/pixel-measurement/jdkcdajnaldgjmkdkkkgenbgdajaaapa


Universal Design